Government of Karnataka
Department of Technical Education

Diploma in Electrical & Electronics Engineering

C-25 Scheme of Studies

(Effect from the AY 2025-26)

Government of Karnataka

DEPARTMENT OF TECHNICAL EDUCATION

Curriculum Structure

I Semester Scheme of Studies

Sl. No.	Teaching Department	Course Code	Course Name	Hours per week			Total Contact Hours /week	Credits	CIE Marks		Theory SEE Marks		Practice SEE Marks		Total Marks
				L	T	P			Max	Min	Max	Min	Max	Min	
Integrated Courses															
1	SC	25SC11I	Engineering Mathematics-I	4	0	4	8	6	50	20	50	20	-	-	100
2	ME	25ME02I	Computer Aided Engineering. Graphics	3	0	4	7	5	50	20	-	-	50	20	100
3	EE	25EE01I	Fundamentals of Electrical & Electronics Engineering.	3	0	4	7	5	50	20	-	-	50	20	100
4	EE	25EE11I	Basic Electrical Technology	4	0	4	8	6	50	20	50	20	-	-	100
Audit Course															
5	EE	25EE12T	Environmental Sustainability	2	0	0	2	2	50	20	-	-	-	-	50
6	Personality Development		NCC/NSS/YOGA/SPORTS…	Students are expected to engage in any one of these activities from 1st semester to 6th semester(No Credits)											
			Total	**16**	**0**	**16**	**32**	**24**	**250**	**-**	**100**	**-**	**100**	**-**	**450**

Government of Karnataka
DEPARTMENT OFTECHNICAL EDUCATION

Curriculum Structure

II Semester Scheme of Studies

Sl. No.	Teaching Department	Course Code	Course Name	Hours per week			Total Contact Hours /week	Credits	CIE Marks		Theory SEE Marks		Practice SEE Marks		Total Marks
				L	T	P			Max	Min	Max	Min	Max	Min	
Integrated Courses															
1	SC	25SC21I	Engineering Mathematics-II	4	0	4	8	6	50	20	50	20	-	-	100
2	CS	25CS01I	IT Skills	3	0	4	7	5	50	20	-	-	50	20	100
3	ENG	25EG01I	Essential English Communication	4	0	4	8	6	50	20	-	-	50	20	100
4	EE	25EE21I	Concepts of Electrical Circuit	4	0	4	8	6	50	20	50	20	-	-	100
Audit Course															
5	EE	25EE22T	Indian Constitution	2	0	0	2	2	50	20	-	-	-	-	50
6	Personality Development		NCC/NSS/YOGA/SPORTS…	Students are expected to engage in any one of these activities from 1st semester to 6th semester (No Credits)											
			Total	**16**	**0**	**16**	**32**	**25**	**250**	**-**	**100**	**-**	**100**	**-**	**450**

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III Semester Scheme of Studies

Sl. No.	Teaching Department	Course Code	Course Name	Hours per week			Total Contact Hours /week	Credits	CIE Marks		Theory SEE Marks		Practice SEE Marks		Total Marks
				L	T	P			Max	Min	Max	Min	Max	Min	
Integrated Courses															
1	EE	**25EE31I**	Generation, Transmission And Distribution	4	0	4	8	6	50	20	50	20	-	-	100
2	EE	**25EE32I**	Switchgear and Protection	4	0	4	8	6	50	20	50	20	-	-	100
3	EE	**20EE33I**	Generators & Transformers	3	0	4	7	5	50	20	-	-	50	20	100
4	EE	**25EE34I**	Analog and Digital Electronics	3	0	4	7	5	50	20	-	-	50	20	100
Audit Course															
5	KAN	25KA31T	Kannada –I ಸಾಹಿತ್ಯ ಸಿಂಚನ-1/ಬಳಕೆ ಕನ್ನಡ-1	2	0	0	2	2	50	20	-	-	-	-	50
			Total	**16**	**0**	**16**	**32**	**24**	**250**	**-**	**100**	**-**	**100**	**-**	**450**

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Curriculum Structure

IV Semester Scheme of Studies

Sl. No.	Teaching Department	Course Code	Course Name	Hours per week			Total Contact Hours /week	Credits	CIE Marks		Theory SEE Marks		Practice SEE Marks		Total Marks
				L	T	P			Max	Min	Max	Min	Max	Min	
Integrated Courses															
1	EE	**25EE41I**	Electrical Motors & Drives	4	0	4	8	6	50	20	50	20	-	-	100
2	EE	**25EE42I**	Power Electronics	4	0	4	8	6	50	20	50	20	-	-	100
3	EE	**25EE43I**	Fundamentals Of Automation Technology	3	0	4	7	5	50	20	-	-	50	20	100
4	EE	**25EE44I**	Electrical Power System Design	3	0	4	7	5	50	20	-	-	50	20	100
Audit Course															
5	KAN	25KA41T	Kannada –II ಸಾಹಿತ್ಯ ಸಿಂಚನ-II/ಬಳಕೆ ಕನ್ನಡ-II	2	0	0	2	2	50	20	-	-	-	-	50
			Total	**16**	**0**	**16**	**32**	**24**	**250**	**-**	**100**	**-**	**100**	**-**	**450**

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	Electrical & Electronics Engineering	**Semester**	**I / II**
Course Name	Fundamentals of Electrical and Electronics Engineering	**Type of Course**	Integrated
Course Code	**25EE01I**	**Contact Hours**	7 hours/week 91 hours/semester
Teaching Scheme	L:T:P::3:0:4	**Credits**	5
CIE Marks	**50**	**SEE Marks**	**50 (Practice)**

1. Rationale:

Fundamentals of Electrical and Electronics Engineering is essential for all streams of diploma engineering. At the end of this Course the student is able to gain knowledge about electrical safety, DC and AC circuits, Electrical wiring circuits, protective devices, electrical machines, and working of electronics devices

2. Course Outcomes: At the end of the Course, the student will be able to:

CO-01	Comply with the Electrical safety procedures
CO-02	Analyze DC circuits, AC circuits and electrical wiring systems
CO-03	Understand electrical machines
CO-04	Evaluate Battery and UPS requirements
CO-05	Identify and analyze the working of Electronic components

3. Course Content

WEEK	CO	PO	Theory	Practice
1	1	1,4	Electrical safety Procedures: ● Meaning of Electrical Safety ● Safety precautions in electrical working place ● Electrocution (Electric shock) and How to free a person from electrocution.	● Demonstrate use of Personal Protective Equipment (PPE) and types ● Electrocution (Electric shock) Use Videos to demonstrate how to free a person from electrocution
	1	1,4	First aid in Electrical injury and methods Electrical fire, causes and preventions Fire extinguishers and types	Demonstration of Pipe and plate Earthing methods Know your Electrical lab. Identify power supply, various components with symbols, Check earthing by measuring the voltage between neutral
	1	1,4	Types of Electrician Tools and their functions Earthing:	

			Definition, necessity and types Advantages of Earthing and applications	and earth points.
2	2	1,4	Sources of Electricity ● Sources of Electricity- Conventional and Non-conventional sources ● Advantages of electrical energy ● Effects of electric current and its applications	Video demonstration on identification and observation of different ranges and types of meters Verification of Ohm's Law using simple circuit
	2	1,4	Definition, units and meters; Electric Current, Voltage, Resistance, Potential Difference, EMF Ohm's Law; ● Statement, explanation, Applications and limitations. ● Simple Problems on Ohm's law.	Demonstrate experimentally Open circuit, closed circuit and short circuit conditions in Simple series circuit.
	2	1,4	Circuit conditions- open, close, and short circuit, definitions of series, parallel and series-parallel	
3	2	1,4	Derivation of Effective resistance and total current, voltages calculations in Series and Parallel circuits Applications of above circuits	Determine the equivalent Resistance in series circuit
	2	1,4	Simple problems on Series circuit	
	2	1,4	Simple problems Parallel circuit. Calculations of equivalent resistance for simple series, parallel and series-parallel circuits	Determine the equivalent Resistance in parallel circuit
4	2	1,4	Definitions, units and meters; Electrical work, power and energy, Simple problems on Electrical energy consumption (Unit/ KWh)	Measure the AC voltage, current, power using relevant measuring instruments in a Single phase AC circuit Or Measurement of energy in a Single phase AC circuit
	2	1,4	**AC Fundamentals**; Sinusoidal voltage, current, definitions of amplitude, cycle, time-period, frequency, phase, phase difference	Demonstrate the measurement of Amplitude, peak-peak value, time period and frequency of AC quantity using CRO and function generator.
	2	1,4	Single phase and Three phase electrical power supplies Applications of single-phase and three-phase power Advantages and Differences between single-phase and three-phase power.	
5	2	1,4	**Protective Devices** • Necessity of protective devices, functions and Applications; ✔ Rewireable Fuse ✔ Glass cartridge fuse ✔ HRC fuse	Wire up and test PVC Conduit wiring to control two lamps and one socket independently by providing suitable protective devices. Check the avail;ability of supply and polarities using a standard tester.
	2	1,4	✔ Kit-kat fuse	Wire up and test to control one lamp

			✔ MCB ✔ ELCB Video demonstration on working of above protective devices	from two places using suitable protective devices (Two- way control/ Staircase wiring) Or Wire up Panel board wiring with protective devices (fuse, MCB, load, Neutral link)
	2 2	1,4	Types of wiring systems and accessories	
6	3	1,4	Electromagnetic Induction: Definition, Faraday's laws, Statically and dynamically induced EMF generation. Self and mutual induced EMF.	Control a lamp using Electromagnetic Relay Or Demonstrate the concept of Electromagnetic induction using coil, magnet and galvanometer
	3 3	1,4	**Transformers:** Function, working, video demonstration on construction of transformer	Determine experimentally the transformation ratio of a given Transformer
	3	1,4	Types and applications, Transformation ratio.	
7	3	1,4	**Generators**: DC and AC Generators- definition, types and applications	Study the Name plate details of a given Electrical machine
	3	1,4	**DC Motors:** Definition, types and applications, BLDC motor	Video demonstration of construction of three phase Induction Motor. Wire up a starter to start and reverse the 1 Phase/3 phase AC motor.
	3	1,4	**AC Motors:** Definition ,Types & Applications Necessity of starters for AC motors, Types and applications.	
8	4	1,4	**Cells and Batteries:** Definition, symbol, types, comparison and applications	Construct a simple battery using primary cells for the required voltage and to lit an LED through a resistor in series.
	4	1,4	Lead Acid battery and its maintenance	Video demonstration on construction of Lithium-Ion battery
	4	1,4	Lithium-Ion battery and its applications	
9	4	1,4	Ampere-Hour Capacity, Selection criteria of batteries	Prepare a report by carrying out periodic maintenance works required for batteries at institutional level.
	4	1,4	**UPS** • Meaning, types and applications	
	4	1,4	Block diagram of UPS system.	Connect UPS in an electrical lighting system and observe continuity of supply.
10	4	1,4	Introduction of Alternate Energy Sources and Applications	Video Demonstration on working of a Simple Electric Vehicle
	4	1,4	Evolution of Electric Vehicles, Batteries used for EVs	
	4	1,4	Electric Motors used in EVs Battery and UPS ratings for Solar powered Street lighting	Video demonstration on solar powered street lighting
11	5	1,4	**Electronic Components** Resistors, Capacitors and Inductors - Definition, Unit, Types, and Applications.	Identify and determine the value of resistance, inductance and capacitance using digital multi meter/LCR meter Compute the value of a given carbon resistor using colour coding.

	5	1,4	Definitions of conductors, insulators and semiconductors with examples comparison and applications	Identify the terminals of Diode. Connect the diode in forward and reverse bias modes and observe the status of LED connected in the circuit.
	5	1,4	Semiconductor Diode; • Definitions of P and N type semiconductors, • Diode and its Symbol • Working of Diode in forward and reverse bias • Types of diodes and ratings • Applications	
12	5	1,4	Rectifier • Definition, types, working of Bridge rectifier	Trace the input and output waveforms of an IC Bridge rectifier.
	5	1,4	Transistor • Definition and Types • Symbol • Applications	
	5	1,4	**Digital fundamentals** Integrated Circuits (IC) - Definition, advantages and applications Logic Gates- Symbol, Boolean expression and truth table of AND, OR, NOT, NAND,NOR, EX-OR gates	Verify the truth tables of AND, OR, NOT, NAND, NOR, EX-OR gates
13	5	1,4	**Sensors and Actuators;** • Definition, Types and Applications of sensors	Construct Photo-diode circuit and Test for its working Or Detect an object using IR proximity sensor
	5	1,4	LDR, Photodiode, and Photo transistor (opto-isolator), Solar cell- Symbol and Applications	
	5	1,4	Definition of Actuator, Types and applications. PLC- Block diagram and PLC applications.	Video demonstration on working of any simple Mechanical/ Electrical Actuator. List commercially available PLCs.

4.References:

1. ABC of Electrical Engineering by B. L. Theraja and A. K. Theraja, S Chand Pu
2. Basic Electrical and Electronics Engineering by S. K. Bhattacharya, Pearson Education India, 2012 Edition
3. Electronic Devices and Circuits by I. J. Nagrath, PHI Learning Pvt. Ltd., 2007 Edition.
4. Basic Electrical Engineering byV. Mittle and ArvindMittle, McGrawHill Companies, 2005 Edition
5. The 8051 Microcontroller & Embedded systemsusinkbnnnjbbh bb vvvvg assembly and C (2ndEdition)– M.A.Mazidi , J.C. Mazidi&R.D.McKinlay ISBN: 81-317-1026-2
6. Programmable Logic controllers, W BOLTON
7. https://www.youtube.com/watch?v=mc979OhitAg&list=PLWv9VM947MKi 7yJ0 FCfzTBXpQUQd3K
8. **https://www.youtube.com/watch?v=CWulQ1ZSE3c**
9. en.wikipedia.org/wiki/Transformer
10. **http://www.animations.physics.unsw.edu.au//jw/AC.html**
11. **http://www.alpharubicon.com/altenergy/understandingAC.htm**
12. **http://www.electronics-tutorials/**
13. **http://www.pitt.edu/~qiw4/Academic/ME2082/Transistor%20Basics.pdf**
14. **http://www.technologystudent.com/elec1/transis1.htm**

15. http://www.learningaboutelectronics.com/
16. http://www.electrical4u.com/
17. https://www.youtube.com/watch?v=zLW
18. https://www.youtube.com/watch?v=8PTNjw-hQIM

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

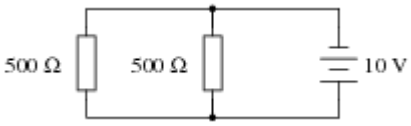
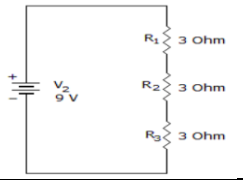
6. CIE – Practice Assessment Methodologies

Sl. No	SEE – Practice Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Practice	180	50	20

7. CIE Theory Test model question paper

Program	Electrical & Electronics Engineering			**Semester -I/II**	
Course Name	Fundamentals of Electrical and Electronics Engineering			**Test**	**I/III**
Course Code	**25EE01I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	a) List any 5 electrician tools. Mention their function. b)State and explain Ohm's Law. List the limitations of Ohm's Law.	1 2	1 2	5 8

	c)Compare single-phase and three-phase power. d)Calculate the total current drawn by the below circuit.  	3 4	2 2	6 6
2	a)Define earthing. List the different types of earthing. b)With simple diagrams explain open circuit, short circuit and closed circuit c)What are the Advantages of three phase power over single phase power. d) Find the equivalent resistance and the current flowing through 3-ohm resistance in the circuit below.  	1 2 3 4	1 2 2 2	5 8 6 6
Section - 2				
3	a)List any 5 sources of electrical energy b) Draw a sinusoidal waveform and name amplitude, instantaneous value, time period, frequency and cycle. c)How to free a person from electrocution? d)An electric fan draws a current of 0.9 Amps when connected to a single-phase, 230 volts, 50 HZ AC supply through an electric regulator. Determine the resistance of the regulator.	1 2 3 4	1 2 2 2	5 8 6 6
4	a)List the safety precautions to be taken in an electrical working place b)Define Current,Voltage and Resistance. Mention their symbol and Units. c)List the different types of fire extinguishers. Which type of fire extinguisher do you use to extinguish fire caused by wood and cooking oil? d)A 9 Volt cell when used in an electrical circuit turns on a LED. State the effect of electric current. Justify	1 2 3 4	1 2 2 2	5 8 6 6
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. CIE Practice Test model question paper

Program	Electrical & Electronics Engineering			**Semester**	**I/II**
Course Name	Fundamentals of Electrical and Electronics Engineering			**Test**	**II/IV**
Course Code	**25EE01I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					

Questions	**CO**	**Marks**
Verify Ohm's Law using a simple circuit.	**1**	**50**
OR Demonstrate experimentally Open circuit, closed circuit and short circuit conditions in Simple series circuit.	**2**	
OR Determine the equivalent Resistance in series circuit	**2**	
OR Determine the equivalent Resistance in parallel circuit		
OR Measure the AC voltage, current, power using relevant measuring instruments in a Single phase AC circuit	**2**	
OR Measurement of energy in a Single phase AC circuit	**2**	
OR Measure Amplitude, peak-peak value, time period and frequency of AC quantity using CRO and function generator.	**2**	
OR Wire up and test PVC Conduit wiring to control two lamps and one socket independently by providing suitable protective devices.	**2**	
OR Wire up and test to control one lamp from two places using suitable protective devices (Two- way control/ Staircase wiring)	**2**	
Or Wire up Panel board wiring with protective devices (fuse, MCB, load, Neutral link)	**2**	
OR Control a lamp using Electromagnetic Relay	**2**	
Or Demonstrate the concept of Electromagnetic induction using coil, magnet and galvanometer	**3**	
OR Determine experimentally the transformation ratio of a given Transformer	**3**	
OR Wire up a starter to start and reverse the 1 Phase/3 phase AC motor.	**3**	
Scheme of assessment		
a) Identification of meters/ equipment/wires/tools etc		**5**
b) Writing Circuit/writing diagram and Procedure		**12**
c) Conduction		**18**
d) Results		**5**
e) Viva-voce		**10**
	Total Marks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	Using suitable meters/ instruments give the practical working circuits to measure Resistance, Current, Voltage, Power and Energy in DC and AC (Single phase) Circuits.
02	List out the different types of wiring systems used in your laboratories or house with their representation
03	Mini-Projects: Like preparing extension box, switch box and wiring models
04	List out the different protective devices used in your laboratories or house with their ratings.
05	Applications of Electro Magnetic Induction, statically induced and dynamically induced emf, self and mutual induced emfs
06	Prepare a report on types of starters and enclosures used for various industrial applications of AC motors.
07	Types of Cells and Battery maintenance
08	Visit nearby Battery charging shop or show room and prepare a report of the visit.
09	Prepare a report on various types of diodes used for various industrial applications.
10	Prepare a report on various types of sensors and actuators used for various industrial applications.
11	Mini-Projects: Connect and test a sensor (domain application) to a Digital circuit
12	Prepare a report stating HP rating, types and applications of FHP motors.

9. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

10. SEE- Model Practice Question Paper

Program	Electrical & Electronics Engineering		**Semester**	
Course Name	Fundamentals of Electrical and Electronics Engineering	**Course Code 25EE01I**	**Duration**	**180 min**

Questions	**CO**	**Marks**
Verify Ohm's Law using a simple circuit.	**1**	**50**
OR		
Demonstrate experimentally Open circuit, closed circuit and short circuit conditions in Simple series circuit.	**2**	
OR		
Determine the equivalent Resistance in series circuit		
OR	**2**	
Determine the equivalent Resistance in parallel circuit		
OR		
Measure the AC voltage, current, power using relevant measuring instruments in a Single phase AC circuit	**2**	
OR		
Measurement of energy in a Single phase AC circuit		
OR	**2**	
Measure Amplitude, peak-peak value, time period and frequency of AC quantity using CRO and function generator.	**2**	
OR		
Wire up and test PVC Conduit wiring to control two lamps and one socket independently by providing suitable protective devices.	**2**	
OR		
Wire up and test to control one lamp from two places using suitable protective devices (Two- way control/ Staircase wiring)	**2**	
Or		
Wire up Panel board wiring with protective devices (fuse, MCB, load, Neutral link)		
OR	**2**	
Control a lamp using Electromagnetic Relay		
Or	**2**	
Demonstrate the concept of Electromagnetic induction using coil, magnet and galvanometer	**3**	
OR		
Determine experimentally the transformation ratio of a given Transformer		
OR	**3**	
Wire up a starter to start and reverse the 1 Phase/3 phase AC motor.	**3**	
OR		
Construct a simple battery using primary cells for the required voltage and to lit an LED through a resistor in series.	**4**	
OR		
Identify and determine the value of resistance, inductance and capacitance using digital multi meter/LCR meter	**5**	
OR	**5**	
Compute the value of a given carbon resistor using colour coding.		
OR		
Connect the diode in forward and reverse bias modes and observe the status of LED connected in the circuit.	**5**	
OR	**5**	
Trace the input and output waveforms of an IC Bridge rectifier.		
OR	**5**	
Verify the truth tables of AND, OR, NOT, NAND, NOR, EX-OR gates (ANY 3)	**5**	
OR		

Construct Photo-diode circuit and Test for its working Or Detect an object using IR proximity sensor	**5**	
Scheme of assessment **a) Identification of meters/ equipment/wires/tools etc** **b) Writing Circuit/writing diagram and Procedure** **c) Conduction** **d) Results** **e) Viva-voce**		**5** **12** **18** **5** **10**
	Total Marks	**50**

1) Signature of the Examiner **2) Signature of the Examiner**

11. Equipment/softwarelistwithSpecificationforabatchof30students

	Particulars	Specification	Quantity
01	Dual Channel ,continuously variable DC Regulated Power Supply with Current and Overload Protection	30 V, 2 A	05
02	Fixed DC Regulated Power Supply	+/- 15 V, 2 A	05
03	Portable Moving Coil DC Voltmeters	a) 0 - 1 V b) 0 - 10 V c) 0 - 30 V	Each 05 No.
04	Portable Moving Iron AC Voltmeters	a) 0 - 300 V b) 0 - 600 V	Each 05 No.
05	Portable Moving Coil DC Ammeters	a) 0 - 100 mA b) 0 - 1 A c) 0 - 2 A	Each 05 No.
06	Portable Moving Iron AC Ammeters	a) 0 - 2 A b) 0 - 5 A c) 0 - 10 A	Each 05 No.
07	Watt-meters	a) 150/ 300V, 2 A, UPF b) 300/ 600 V, 5/ 10 A, LPF	Each 02 No.
08	Rheostats	25 Ohms, 50 Ohms, 150 Ohms, 220 Ohms (all rated at 3 A)	Each 05 No.
09	Rheostat Loads	1 KW, 230 V	02
10	Wire wound Resistors	-5 Ohms 2 Watts, 25 Ohms 5 Watts, 330 Ohms 2 Watts, 560 Ohms 2 Watts, etc.	Each 05 No.
11	Soldering Iron	60 W	05
12	Single Phase Energy meter	10 A, 230 V, 50 Hz, Digital type	05
13	Multi-meter Digital	¾”	06
14	Dual Trace Oscilloscope	30 MHz	02
15	Three Phase Induction Motor	1 HP – 440 V 50 Hz, 2 HP – 440 V 50 Hz.	Each 02 No.
16	Three phase DOL, Star-Delta, Auto transformer starter		Each 02 No.
17	UPS	1 KVA	01
18	Battery and Hydrometers	Battery Lead-Acid type, 140 A-hr	02
19	I C Trainer kit		05

20	Digital IC's	7400, 7402, 7404, 7408, 7486 etc	Each 10 No
21	Wooden Wiring board	(2x3) ft	10
22	**Wiring accessories** a) PVC conduit - ¾" - 10 lengths b) Cap and casing - ¾" - 10 lengths c) Switches Single Pole- 5A, 230 V d) Switches two way – 5 A, 230 V e) 3 Pin Sockets 5A, 230 V f) Bulb Holders – 5 A, 230 V g) 3 Pin Plug 5A, 230 V h) 60 Watts Lamps i) 100 Watts Lamps j) 15 W CFL lamps k) Copper Wires of sizes 1.5 mm2, 2.5 mm2, 4 mm2 – 1 coil each l) Gang boxes (1+1, 2+1, 2+2) m) Kit –Kat fuses 5A, 15 A n) MCB 16 A & 32 A/ 230 V, Single and Double Pole o) ELCB 16 A & 32 A/ 230 V, Double Pole p) Neutral link- 16 A, 230 V q) Screws of assorted sizes r) Testers		Each 10 No
22	**Electronic Components** a) Diodes - BY 127 and IN 4001 b) Zener Diodes – 6.2 V, 5.6 V, 7.8 V c) Relays – solid state Sugar cube type, SPST, Coil 6V, Power circuit 230 V, 5 A. d) Spring Boards e) Bread Boards f) Tag Boards.		Each 10 No
23	Simple PANEL BOARD/ CUBICAL consisting of bus-bars, CB/MCB/ELCB, meters, HRC fuses, magnetic contactors, cables, earthing points.		01

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	ELECTRICAL AND ELECTRONICS ENGINEERING	**Semester**	**I**
Course Name	BASIC ELECTRICAL TECHNOLOGY	**Type of Course**	Integrated
Course Code	**25EE11I**	**Contact Hours**	8 hours/week 104 hours/semester
Teaching Scheme	L: T:P-4-0-4	**Credits**	6
CIE Marks	**50**	**SEE Marks**	**50 (Theory)**

1. Rationale:

The inclusion of these topics in a Basic Electrical Technology course provides a comprehensive foundation for understanding and working with electrical systems. This curriculum is designed to equip students with essential theoretical knowledge and practical skills, ensuring they can safely and effectively design, install, maintain, and troubleshoot electrical systems. By covering both fundamental concepts and hands-on practices, students gain the expertise needed for a variety of electrical technology applications.

2. Course Outcomes : At the end of the Course, the student will be able to:

CO-01	Understand and apply concepts like current, voltage, resistance, power, and solve related circuit problems.
CO-02	Identify and use electrical components (resistors, capacitors, inductors) and solve problems related to their combinations in circuits
CO-03	Implement electrical safety standards and practices, including circuit protection, testing, and hazard identification.
CO-04	Design and wire basic residential lighting circuits, calculate loads, and determine conductor sizes
CO-05	Understand and apply concepts of solar, wind, and hydrogen energy generation, along with fuel cell applications.

3. Course Content

WEEK	CO	PO	Theory	Practice
1	1,2	1,2,3,7	**Basic Electrical Concepts:** **Electricity Basics:** **Current:** Types (AC, DC), units (amperes), and measurement. **Voltage:** Definition, units (volts), and relationship with current and resistance.	Video demonstration of flow of current in a simple circuit under the influence of electric field
	1,2		Definition, unit, :Potential Difference, E.M.F. **Resistance:** Definition, units (ohms), and Factors affecting resistance.	Compute and compare the resistivity for different materials(Copper, Aluminium, Nichrome) by suitable

	1,2		Laws of Resistance, Specific Resistance and its unit. Sample problems	method
	1,2		Effect of temperature on resistance and temperature coefficient..	
2	1,2	1,2,3,7	Series, parallel and series-parallel combination circuits Solve simple problem	**Aim 1:**To measure and calculate the total resistance of resistors connected in series **Aim 2**: To measure and calculate the total resistance of resistors connected in parallel. **Aim 3**: To measure and calculate the total resistance of a mixed series-parallel circuit.
	1,2	1,2,3,7	The relationship between voltage (V), current (I), and resistance (R) is V = I × R. **Power:** Definition, units (watts), and formula (P = V × I). **Energy:** Definition, units (Joules), and energy consumption calculations. Solve simple problems	
	1,2	1,2,3,7	**Electrical Components:** **Resistors:** Types, symbols, color coding, and applications. **Capacitors:** Types, symbols, capacitors in series and parallel- derivation, and applications. **Inductors:** Types, symbols, and applications ● Solve simple problem - equivalent inductance of the circuit	**Aim 1**: To calculate the effective capacitance of capacitors connected in series and parallel **Aim 2**: To measure the equivalent inductance of inductors connected in series and parallel.
	1,2	1,2,3,7	Solve simple problem on 1.Compute the resistance value of a carbon resistor for a given set of colour bands 2. Compute effective capacitance of capacitors connected in series and parallel combination	
3	1,2,3	1,2,34,7	**Measuring Instruments:** Basic Electrical Measuring Instruments Types,in AC .DC and Applications. Ammeters,Voltmeters,Ohmmeters,Wattmeters, Energy Meters,TriVector Meter	**Aim 1**: To measure current using an ammeter in both AC and DC circuits. **Aim 2**: To measure voltage using a voltmeter in both AC and DC circuits. **Aim 3**: To measure power using a wattmeter in a single-phase AC circuit.
	1,2,3	1,2,34,7	Classification of Measuring Instruments ● **Analog Instruments**: galvanometers, ammeters, voltmeters, and wattmeters.	
	1,2,3	1,2,34,7	● **Digital Instruments**: digital voltmeters, digital multimeters, and digital oscilloscopes.	**Aim 1**: To connect and configure an oscilloscope to measure and display AC and DC waveforms. **Aim 2**: To use the oscilloscope to determine frequency, amplitude, and phase of a sinusoidal signal.
	1,2,3	1,2,34,7	● **Comparison of Analog vs. Digital Instruments**: Advantages, disadvantages, and appropriate usage scenarios. **Oscilloscope:** Basic usage for viewing waveforms **Function Generator:** Generating different types of electrical signals.	
4	3	1,2,3,4,	**Basic Equipment Installation Safety Practices:** **Definition and Importance:** Overview of	**Safety Procedures:** Practicing safe handling of electrical components and instruments.

		7	electrical safety and its importance in preventing accidents and ensuring safe electrical installations.	Prepare a chart showing all safety tools used while working with electrical circuits. Prepare a chart showing standard symbols used in electrical circuits.
	3	1,2,3,4,7	**Basic Electrical Hazards in lighting and power installations:** Understanding common electrical hazards such as electric shock, arc flash, and electrical fires	
	3	1,2,3,4,7	**National and International Standards:** Overview of key safety standards such as IEC, NEC (National Electrical Code), and OSHA regulations	
	3	1,2,3,4,7	Proper use of personal protective equipment (PPE). Understanding electrical hazards and safe working practices	
5	**3**	1,2,3,4,7	**.Circuit Protection:** Fuses, circuit breakers, and their functions	**Circuit Protection:** Video Demonstration on the use of fuses and circuit breakers in protecting circuits. **Aim 1**: To study the construction and working of fuses and measure the current rating of a fuse for a given circuit. **Aim 2**: To understand the working of circuit breakers and demonstrate their tripping mechanism during overload or short-circuit conditions. Operate a given fire extinguisher to put out the fire in a place.
	3	1,2,3,4,7	**Isolation Techniques:** Methods for safely isolating electrical circuits and equipment in a factory/workshop maintenance.	
	3	1,2,3,4,7	**Testing for Voltage:** How to safely test for the presence or absence of voltage using appropriate tools and meters.	
	3	1,2,3,4,7	**Verification of De-Energization:** Procedures to confirm that circuits are de-energized before starting work.	
6	4	1,2,3,4,7	Wiring Systems & Standards ● **Types of Electrical Wiring Systems:** ○ Surface wiring, concealed wiring, conduit wiring, tray wiring, etc. ○ Comparison of various wiring systems.	**Adopted Basic Wiring Techniques** ● Wire Stripping and Preparation: ○ Tools and techniques for stripping insulation from cables. ● Crimping and Soldering: ○ Making reliable wire connections using crimping tools or soldering. ● 1 sq mm Cable Terminations and lugging: ○ Correct techniques for terminating wires with connectors or terminal blocks. ● Using Hand Tools: ○ Proper use of wire cutters, pliers, screwdrivers, tape measures, etc. Carry out proper techniques for
	4	1,2,3,4,7	Types of wires and cables, components and accessories of electrical wiring systems**.**	
	4	1,2,3,4,7	**Voltage Levels:** ● Low voltage, medium voltage, and high voltage applications.	
	4	1,2,3,4,7	**Color Codes and Wire Identification:** ● Standards for wire color coding to distinguish live, neutral, and ground wires.	

				stripping insulation and terminate one sq mm cable.
7	4	1,2,3,4,7	**Wiring Installations** ● **Residential Wiring:** ○ Basic circuits for lighting and power distribution. ○ Single-phase and three-phase circuits	● **Aim**: To design and install a functional lighting circuit for a single room to meet a practical load requirement (e.g., controlling two LED lights and one ceiling fan). ● Practice Cable laying, Cable joints (terminations),proper size lugs, crimping of joints towards a house wiring installation with Wire joints
	4	1,2,3,4,7	● List code of Practice for Lighting Installations ● Point out the factors to be considered while selecting the type of wiring system	
	4	1,2,3,4,7	● List the materials required for Interior wiring. Write their specifications. ● Specify the method of deciding the number of sub-circuits,	
	4	1,2,3,4,7	● Solve one simple example on Interior wiring with a single circuit- calculate the load-for a residential building. ● Determine the size of conductors	
8	4	1,2,3,4,7	● Solve one simple example on Interior wiring with a single circuit- calculate the load-for a residential building. ● Determine the size of conductors,	● Wire up a simple circuit consisting of a Electrical Wiring types (godown) ● Wire up for a calling bell and troubleshoot if any.
	4	1,2,3,4,7	● Prepare the schedule of materials for providing lighting circuit wiring for the above	
	4	1,2,3,4,7	● Wire up basic lighting circuits with switches and fixtures. (Draw wiring plan for the above residential building)	
	4	1,2,3,4,6,7	● Concept of Phase wire, Neutral wire, Earth wire and Half wire	
9	4	1,2,3,4,6,7	● Troubleshoot and work on simple lamp Circuits for different scenarios, Example on Single-Phase Circuit to control 2 lamps in series(Lighting Circuit)	One lamp, one fan and one three pin socket controlled from one switchboard One lamp controlled from three different places. (Intermediate switch) with protective devices. Construct the main distribution board with SP, DP switches, protective devices and a lamp load.
	4	1,2,3,4,6,7	● Learn about Ground Fault Circuit Interrupter (GFCI) outlets and their importance in preventing electrical shocks.	
	4	1,2,3,4,7	● Different types of switches commercially available in the market with their rating and specification.	
	4	1,2,3,4,7	● Understand how two way switches control lighting from different locations.	
			Relays - Meaning, types, working of Electromagnetic Relay	Connect a relay to turn on a lamp on its actuation.
			Meaning of electrical contactors- working	

10	4	1,2,3,4,7	and applications	
			Timers- Meaning, types, working of ON timer with example	Connect a timer with a contactor to run a small water pump for a fixed time(ten minutes).
			Working of OFF timer and Delay timer with example	
11	4	1,2,3,4,7	Testing of electrical wiring ● Continuity Testing	Testing domestic wiring installation: • Continuity test (OC & SC Test) • Polarity test Earth and ground test • Insulation and leakage test
	4	1,2,3,4,7	● Insulation Resistance Testing	
	4	1,2,3,4,7	● Grounding Resistance Testing	
	4	1,2,3,4,7	● polarity of control and protective devices,	
12	4	1,2,3,4,7	● Preventive maintenance practices for electrical systems	Design a 2BHK residential installation scheme and estimate the materials required. Draw the layout diagram; installation plan and wiring diagram according to National electrical code for the following wiring installations.Also prepare the bill of materials.
	4	1,2,3,4,7	Service mains ● Meaning of service mains, code of Practice for service mains, types of service mains	
	4	1,2,3,4,7	● Overhead Service Mains -materials and specifications ● UG Service Mains -materials and specifications	
	4	1,2,3,4,7	● schedules of materials for single phase OH service connection,	Consumer main board with Energy meter, Double pole iron Clad Switch and Distribution Board
13	5	1,2,3,4,7	● **Alternate Energy Sources** ● List different types of alternative energy sources. ● Components used in Solar Power Plants. ● Discuss generation of electrical energy using solar energy with a block diagram.	Run a miniature solar powered fan.
	5	1,2,3,4,7	● Components used in Wind Power Plants ● Discuss generation of electrical energy using wind energy with a block diagram.	
	5	1,2,3,4,7	● Generation of electricity using hydrogen energy	Provide a solar light to the lawn of your institution.
	5	1,2,3,4,7	● Concept of fuel cells and applications.	

4. References:

1. Electrical Engineering Drawing-Bhattacharya S. K.- New Age International, New Delhi, ISBN: 978-81-224-0855-3.
2. Electrical Wiring, Estimating and Costing - Uppal S.L; Garg G.C. - Khanna Publishers, New Delhi,
3. Electrical Workshop: Safety, commissioning, maintenance and testing of electrical equipment - I.K. International Publishing House , Pvt. Ltd. New Delhi, ISBN:978- 9389447057

4. Electrical Estimating and Costing - Gupta J. B. - S. K. Kataria & Sons, New Delhi,
5. An introduction to electrical engineering - materials Indulkar C.S. & Thiruvengadam S. -S Chand & Co., ISBN :978-8121906661

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2 Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4 Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

SEE - Theory Assessment Methodologies

Sl. No	SEE – Theory Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Theory	90	50	20

6. CIE Theory Test model question paper

Program	ELECTRICAL AND ELECTRONICS ENGINEERING			**Semester - I**	
Course Name	BASIC ELECTRICAL TECHNOLOGY			**Test**	**I/III**
Course Code	**25EE11I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					
Q.No	**Questions**		**Cognitive Level**	**Course Outcome**	**Marks**
Section - 1					
1	a) Describe how voltage, current, and resistance are related in a circuit. b) draw circuit a 12V battery connected to two resistors in parallel: R1=6 Ω and R2=3 Ω. Calculate the total resistance, total current, and current through		L1/L2 L3/L4		

	each resistor. c) draw circuit a 12V battery connected to a series circuit of two resistors, R1=2 Ω and R2=4 Ω. Calculate the total resistance, current, and voltage across each resistor. **d) Energy,Power:** Definition, units (watts) **e) Calculate Energy Consumption of Multiple Appliances** ● **Appliance 1**: A 2000W microwave used for 0.5 hours. ● **Appliance 2**: A 100W bulb used for 4 hours. ● **Appliance 3**: A 1500W air conditioner used for 3 hours. Calculate: 1. The total energy consumed by all appliances in kilowatt-hours (kWh). 2. The total energy consumed in joules (J).	L3/L4 L1/L2 L3/L4	1,2	5+5+5+5+5=25
2	a) **Resistors, Capacitors, Inductors:** Types, symbols b) A resistor has the following color bands: Red, Yellow, Green, and Gold. What is the resistance value of the resistor, and what is its tolerance? c) Calculate the total capacitance of three capacitors connected in series: ● Capacitor C1=10 μF ● Capacitor C2=15 μF ● Capacitor C3=20 μF d) **Transistors:** Basic concepts of BJTs and FETs e) A diode has a forward voltage drop of 0.7V, and an LED has a forward voltage drop of 2V. If they are connected in series with a 12V power supply, calculate: 1. The total voltage drop across the diode and LED. 2. The current flowing through the circuit if the series resistor has a resistance of 330Ω.	L1/L2 L3/L4 L3/L4 L1/L2 L3/L4	1,2	5+5+5+5+5=25
Section - 2				
3	a) Basic Electrical Measuring Instruments Types, and Applications. b) **Analog Instruments**: Working principles and examples like galvanometers, ammeters c) **Digital Instruments**: Working principles and examples like digital voltmeters, digital multimeters, and digital oscilloscopes. d) **Comparison of Analog vs. Digital Instruments** **e) Function Generator:** Generating different types of electrical signals	L1/L2 L3/L4 L1/L2 L3/L4 L3/L4	1,2	5+5+5+5+5=25
4	a) Importance of Electrical Safety b) Difene Electric Shock, Causes and Prevention c) Describe the significance of the National Electrical Code (NEC) in the United States. d) Proper use of personal protective equipment (PPE). e) Understanding electrical hazards and safe working practices	L1/L2 L3/L4 L1/L2 L3/L4 L3/L4	1,2	5+5+5+5+5=25

Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.	

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. CIE Practice Test model question paper

Program	ELECTRICAL AND ELECTRONICS ENGINEERING			**Semester**	
Course Name	BASIC ELECTRICAL TECHNOLOGY			**Test**	**II/IV**
Course Code	**25EE11I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					

Questions	**CO**	**Marks**
1) To measure and calculate the total resistance of resistors connected in series		**50**
2) To measure and calculate the total resistance of resistors connected in parallel.	1,2	
3) To measure and calculate the total resistance of a mixed series-parallel circuit.	1,2	
4) To calculate the effective capacitance of capacitors connected in series and parallel	1,2	
5) To measure the equivalent inductance of inductors connected in series and parallel.	1,2	
6) To measure current using an ammeter in both AC and DC circuits.	1,2	
7) To measure voltage using a voltmeter in both AC and DC circuits.	1,2	
8) To measure power using a wattmeter in a single-phase AC circuit.	1,2	
9) To connect and configure an oscilloscope to measure and display AC and DC waveforms.	1,2	
10) To use the oscilloscope to determine frequency, amplitude, and phase of a sinusoidal signal.	1,2	
11) Practicing safe handling of electrical components and instruments.	3	
12) Video Demonstration on the use of fuses and circuit breakers in protecting circuits.	3	
13) To study the construction and working of fuses and measure the current rating of a fuse for a given circuit.	3	
14) To understand the working of circuit breakers and demonstrate their tripping mechanism during overload or short-circuit conditions.	3	
15) Learn the correct technique for terminating wires using lugs and terminal blocks.	4	
16) Learn how to make reliable electrical connections using crimping tools and soldering.	4	
17) To design and install a functional lighting circuit for a single room to meet a practical load requirement (e.g., controlling two LED lights and one ceiling fan).	4	
18) Practice Cable laying, Cable joints (terminations),proper size lugs, crimping of joints towards a house wiring installation with Wire joints	4	
19)		
Scheme of assessment a) Circuit/Setup Design (10 Marks) b) Practical Execution (10 Marks) c) Correct Connections (10 Marks)		

d) Functionality Testing (10 Marks) e) Safety and vivo (10 Marks)	
TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	Prepare a chart showing different electronic components.
02	Project: Designing and Wiring a Residential Lighting Circuit **Objective:** Design and implement a basic residential lighting circuit for a single room, considering load calculation, safety standards, and energy efficiency.
03	Report: Renewable Energy Systems for Domestic Applications **Objective:** Analyze the feasibility and implementation of renewable energy systems, focusing on solar and wind power, for domestic use
04	Report: Comparative Analysis of Analog and Digital Measuring Instruments **Objective:** Study the advantages, limitations, and applications of analog and digital instruments.
05	Report Template: Renewable Energy Systems for Domestic Applications
06	**Model Making:** ● Build a small-scale solar or wind power generation model.

9. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

10. SEE–Model Theory Question Paper

Program	ELECTRICAL AND ELECTRONICS ENGINEERING	**Semester**	**I**
Course Name	BASIC ELECTRICAL TECHNOLOGY	**Marks**	**50**
Course Code	**25EE11I**	**Duration**	**90 Min**

Note: Answer any one full question from each section. Each full question carries equal marks.

Q No	Questions	Cognitive Levels	Course Outcomes	Marks
	Section -1			
1	a) State Ohm's Law and derive the relationship between voltage, current, and resistance. b) A circuit has a 12V battery and resistors R1=6Ω, R2=4Ω connected in series. Calculate the total resistance and current in the circuit.	L1/L2 L3/L4	1,2 1,2	5 5
2	a) Differentiate between series and parallel circuits. Provide examples of each and their applications. b) Calculate the total resistance and current in a circuit with a 12V battery and resistors R1=5Ω, R2=10Ω, connected in parallel.	L1/L2 L3/L4	1,2 1,2	5 5
	Section -2			
3	a) Describe the different types of resistors and their applications in circuits. b) What are diodes? Explain their working principle and applications in rectification	L1/L2 L3/L4	1,2 1,2	5 5
4	a) Explain the function of capacitors in electrical circuits and outline their various types b) Discuss the types of inductors and their applications in electronic circuits	L1/L2 L3/L4	1,2 1,2	5 5
	Section -3			
5	a) Define electrical safety and its importance in preventing accidents in electrical	L1/L2	3	5

	installations b) Comparison of Analog vs. Digital Instruments	L3/L4	3	5
6	a) Importance of electrical safety and its importance in b) Given a capacitor that stores 5mC of charge at 10V, calculate its capacitance. Explain the significance of capacitance in AC circuits.	L1/L2 L3/L4	3 3	5 5
Section -4				
7	a) Explain the difference between **surface wiring** and **concealed wiring** systems. b) Apply the **Code of Practice** for deciding the number of sub-circuits in a residential installation. For a room measuring 5m x 6m, determine the number of sub-circuits needed and explain your reasoning	L1/L2 L3/L4	4 4	4 6
8	a) Explain how a **two-way switch** works to control lighting from two different locations. b) Create a lighting installation plan for a multi-branch circuit, specifying the load and wiring techniques for a room of 4 x 5 meters	L1/L2 L3/L4	4 4	4 6
Section -5				
9	a) Compare the materials and specifications used for overhead and underground service mains. b) Discuss generation of electrical energy using solar energy with a block diagram.	L1/L2 L3/L4	4 4	5 5
10	a) List common electrical problems found in residential wiring systems b) Discuss generation of electrical energy using wind energy with a block diagram.	L1/L2 L3/L4	4 4	5 5

11. Equipment/software list with Specification for batch of 30 students

Sl.No.	Particulars	Specification	Quantity
01	**Power Supply Unit**:	AC/DC adjustable power supply, 0-30V, 5A.	6
02	Digital Multimeter	3 ½ or 4 ½ digit display	10
03	Resistors (Various Ratings)	Values: 10Ω to 1MΩ, 1/4W, 1/2W	-
04	Capacitors (Various Ratings)	Electrolytic, ceramic, 10μF to 1000μF	-

05	**Inductors**:	Inductors with ratings 10mH to 100mH.	
06	Breadboards	Full-size, with 830 tie-points for circuit assembly	15
07	Connecting Wires	Flexible wires with banana plugs, assorted colors.	
08	**Oscilloscope**:	00 MHz Digital Storage Oscilloscope (DSO)	5
09	Function Generator	1 Hz to 2 MHz, sine, square, triangle wave outputs.	
10	**Ammeter and Voltmeter**:	Analog and digital, ranges 0-10A and 0-300V.	5 each
11	**Fuses**:	Glass and ceramic, 1A to 30A ratings.	
12	Circuit Breakers	Miniature Circuit Breakers (MCB) 1P, 2P, 6A to 32A	6
13	Relay Module	5V/12V SPDT relays.	6
14	Fire Extinguisher	ABC dry chemical type, 2kg.	3
15	Wire Strippers	Adjustable wire gauge compatibility (0.5mm² to 6mm²).	-
16	Crimping Tools	Ratchet crimping tool, 0.5mm² to 6mm².	-
17	**Soldering Kit**:	25W soldering iron, solder wire, flux, and stand.	-
18	Hand Tools	Insulated pliers, cutters, screwdrivers.	-
19	Lugs and Terminals	Copper lugs, 1 sq mm to 16 sq mm.	-
20	Lighting Fixtures		-
21	Switches and Boards		-
22	**Distribution Board**:		-
23	Timer and Contactor		-
24	Solar Kit		6

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	**Electrical & Electronics Engineering**	**Semester**	**II**
Course Name	**Concepts of Electrical Circuits**	**Type of Course**	Integrated
Course Code	**24EE21I**	**Contact Hours**	8 hours/week (104 hours/semester)
Teaching Scheme	L: T:P- 4:0:4	**Credits**	6
CIE Marks	**50**	**SEE Marks**	**50 (Theory)**

1. Rationale:
   Concepts of Electrical Circuits is essential for diploma in Electrical & Electronics engineering. It covers the identification of passive elements, their use in electrical circuits, applying laws and theorems to analyse the dc circuits. Basic knowledge of single-phase and three-phase ac circuits are required to understand the concepts of courses in electrical machines, generation, transmission & distribution, utilization, switchgear, protection etc.

2. Course Outcomes : At the end of the Course, the student will be able to:

CO-01	Understand the concept of electrostatics Identify the passive elements, measure and use them in electrical circuits.
CO-02	Apply the concepts of KCL/KVL and network theorems in solving DC circuits.
CO-03	Demonstrate the electromagnetic induction principle and its applications.
CO-04	Analyze single-phase and three-phase AC circuits

3. Course Content

WEEK	CO	PO	Theory	Practice
1	1	1,2, 4	Electrostatics Define ● Electric charge ● Electric field ● Explain laws of electrostatics	Simple experiments on static electricity such as Static flyer, salt and pepper static electricity experiment, electrostatic soap bubble science experiment, bending water with static electricity
			Define ● Electrostatic induction ● Electric flux ● Electric field intensity ● Flux density	
			● Electric potential and potential difference ● Potential gradient	
			Breakdown voltage and dielectric strength	

2	1	1,2, 4	Capacitors Definition, types, and applications ● Capacitance – definition, unit ● Dielectrics, different materials, unit ● Effect of dielectrics on capacitance, dielectric constant.	Verification of effective capacitance by calculation and measurement using LCR meter a) Capacitors in series b) Capacitors in Parallel c)Capacitors in Series Parallel Identification of different types of inductors.
			Connection of Capacitors ● In series, parallel and series parallel ● Simple problems ● Expression for energy stored in a capacitor (No derivation)	
			Inductors ● Define, unit, types and applications, ● Expression for energy stored in an inductor (No derivation)	
			Connection of inductors in series and parallel connections	
3	2	1.2. 4	Secondary Cells and Batteries: ● Faraday's laws of Electrolysis. ● Define cell and Battery ● Difference between Cell and a Battery. (Primary and Secondary)	Connect secondary cells in series to a simple circuit and measure emf and current. Connect secondary cells in parallel to a simple circuit and measure total emf and current.
			● Grouping of cells. ● Problems on grouping of cells. ● Classification of Secondary Batteries Based on their Uses	
			● Ampere-hour efficiency and watt hour efficiency. ● Types and applications of batteries ● Ni-Cd, Li-Ion, Lead-acid, Ni-Mh, Ni-Polymer ● Explain Construction and working of lithium ion battery	Connect secondary cells in Series - Parallel group to a simple circuit and measure emf and current. Test and report the condition of car battery of 12 V rating using hydro meter and cell tester.
			Charging Systems ● Constant-current System ● Constant-voltage System ● Trickle Charging ● Sulphation-Causes and Cure ● Indication of a fully charged and discharged cell	
4	2	1,2, 4	Define: ● Circuit, parameters, Linear circuit, ● Non-linear circuit, Bilateral, Unilateral,	For simple circuit verify KCL and compare with Theoritcal value
			● Electric network, Passive network, Active network, ● Node, Branch, Loop, Mesh, give examples for each.	For simple circuit verify KVL and compare with Theoritcal value

			● State and explain Kirchhoff's current law, Simple problems	
			● State and explain Kirchhoff's Voltage law, Simple problems	
5			● Ideal Voltage, Ideal Current source, conversion from voltage to current and current to voltage conversion.	Video demonstration on converting star to delta delta to star transformation and prepare report
			● Simple problems	
			● Star - delta Transformation ● Problems on Star - delta Transformation	
			● delta - Star Transformation ● Problems on delta - Star Transformation	
6	2	1,2, 4	Network theorems ● State and explain Thevinin's Theorem, ● simple problems	verify experimentally Thevenin's theorem.
			● State and explain Norton Theorem ● simple problems	
			● State and explain Maximum power transfer Theorem.	Verify experimentally Maximum power transfer Theorem.
			● simple problems on the above	
7	3	1,2, 4	Magnetism and Magnetic circuits Define: Magnetic flux, magnetic field, flux density, Magnetic field strength(H), Absolute Permeability, Relative permeability, Mention their units.	Demonstrate measurement of magnetic flux using flux meter.
			Define magnetic circuit, MMF, Reluctance, laws of reluctance, relation between MMF, reluctance and flux, flux density, magnetic leakage, leakage co-efficient, permanence, and mention their units.	
			State the relationship between Flux, flux density, MMF and Reluctance Compare Electric circuit with magnetic circuit	Demonstrate the measurement of permeability of silicon steel.
			Solve simple problems on magnetic circuit	
8	3	1,2, 4	Electro Magnetic Induction Describe magnetic field around a current carrying conductor. State and apply Cork Screw Rule and Right Hand Thumb Rule to find the direction of field around a current carrying conductor	Rig up a circuit to demonstrate Faraday's laws of electromagnetic induction and the field due to a current carrying conductor.
			State and explain the Faraday's laws of Electromagnetic Induction.	
			Solve simple problems on Electromagnetic Induction.	Demonstrate the use Fleming's Right hand rule to find the

			State and explain Fleming's Right Hand Rule. State and explain Lenz's law. Define statically and dynamically induced emfs and their applications.	direction of emf and current induced in a conductor.
9	3	1.2.4	Define self-induced Emf, mutually induced emf differences between Self-induced emf and Mutually induced emf and state their applications.	Demonstrate the concept of self and mutual induced emf.
			Define Self-inductance and Co-efficient of Self-inductance. Define Mutual inductance and Co-efficient of Mutual inductance	
			Expression for the energy stored in a magnetic field	
			Simple simple problemson above	
10	4	1,2,4	A.C. Principles Generation of Single phase AC voltage in a single turn coil. Define: Cycle, Frequency, Amplitude, Time Period and mention their units.	Measurement of AC Voltage by CRO - Amplitude, Peak to Peak Value, Frequency, Time, Period ,
			Maximum value, RMS value, Average value, Form factor and Peak factor of a sinusoidal wave	
			Instantaneous value of Voltage and Current in terms of their maximum value and time. phase and phase difference, problems on above	Measure power in a single phase AC circuit using ammeter, voltmeter, Wattmeter and calculate power, power factor for different loads like mercury vapour lamp/sodium vapour lamp/fluorescent lamp/LED lamp.
			Vectorial representation of AC quantities for lagging and leading quantities, Define true power, apparent power, reactive power and mention their units. Equation for Power, and Power factor in AC circuits	
11	4	1,2,4	Vector Algebra Represent vectors in Rectangular, Trigonometrically and Polar forms.	Measure experimentally the impedance of a series R-L circuit for different frequencies using function generator, CRO and verify it with theoretical impedance value.
			Convert Rectangular form into Polar form and vice--versa Simple problems Perform the following operation in rectangular form- (1) Addition (2) Subtraction Perform the following operations in Polar form (3) Multiplication (4) Division Simple problems	
			Single Phase AC Circuits Current, power and power factor in a pure resistive circuit	Measure experimentally the impedance of a series R-C circuit for different frequencies using

			Analyze the nature of voltage and current in R-L series circuit through wave forms and vector diagrams. Calculate Impedance, Inductive reactance, Current, Power and Power factor of R-L series circuit.	function generator, CRO and verify it with theoretical impedance value.
12	4	1,2, 4	Analyze the nature of voltage and current in R-C series circuit through wave forms and vector diagrams. Calculate Impedance, capacitive reactance, Current, Power and Power factor in R-C series circuit.	Measure experimentally the impedance of a series RLC circuit for different frequencies using function signal generator, CRO and verify it with theoretical impedance value.
			Analyze the nature of voltage and current in R-L-C series circuit through wave forms and vector diagrams. Calculate Impedance, Capacitive reactance and Inductive reactance, Current, Power and Power factor of R-L-C series circuit	
			Explain series resonance. State the condition for resonance with vector diagram and wave forms. Resonant frequency and Q-factor and mention their units	
			Simple problems	
13	4	1,2, 4	Meaning of 3 phase supply, Define phase sequence Comparison between single phase and three phase supply. Advantages of three phase supply over single phase supply	Demonstrate generation of 3 phase voltage using video. Measure three phase power and energy in a three phase ac circuit.
			Connect a 3-ph system in Star and Delta. relationship between Voltage and Current in 3-ph Star and Delta system.	
			Equation for a 3-ph power. Simple problems on 3-ph power calculation using the formula.	Measure 3-ph. Star and delta voltages and currents. Verify with theoretical values.
			Simple problems on above	

4. References:

1. **Electrical Technology volume-1 by B.L. Theraja.**
2. **Principles of Electrical Engineering by B. R. Gupta**
3. **Basic Electrical Engineering by V.K. Mehta &Rohit Mehta.**
4. **Electrical circuits by K.V.N Gowda**
5. **Fundamentals Hand book of Electrical Science, Module 1, Basic Electrical Theory, Department of Energy, U. S. Department of Energy, June 1992.6. https://www.educationcorner.com/static-electricity-experiments/7. https://wonders.physics.wisc.edu/static-electricity/**
8. **http://www.freeengineeringbooks.com/Electrical/Basic-Electrical-Engineering.php**
9. **https://www.youtube.com/watch?v=vwIdZjjd8fo&t=172s**

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

SEE - Theory Assessment Methodologies

Sl. No	SEE – Theory Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Theory	90	50	20

6. CIE Theory Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester -II**	
Course Name	**Concepts of Electrical Circuits**			**Test**	**I**
Course Code	**20EE21I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	a) Define Electric charge,Electric field b) Explain laws of electrostatics c) Define: Electrostatic induction,Electric flux d) A battery consists of 4 cells, each of emf 1.46V. The cells are connected in parallel to an external resistance of 0.525Ω. If the current supplied by the battery is 0.8A, find the internal resistance of each cell. e) Two batteries A and B are connected in parallel and load of 10 Ω is connected across their terminals. A has an e.m.f. of 12 V and an internal resistance of 2 Ω ; B has an e.m.f. of 8 V and an internal resistance of 1 Ω. Use Kirchhoff's laws to determine the values and directions of the	L1 L1 L1 L3 L3	CO1	5 5 5 5 5

	currents flowing in each of the batteries and in the external resistance. Also determine the potential difference across the external resistance.			5
2	a)Define breakdown voltage and dielectric strength. b) Define the terms (i) Electric charge and (ii) Electric field. c) List the types and applications of capacitors. d)A resistance of 10 W is connected in series with two resistances each of 15 W arranged in parallel. What resistance must be shunted across this parallel combination so that the total current taken shall be 1.5 A with 20 V applied? e)A resistance of 5 W is connected in series with a combination of two parallel resistances of 10 W and 20 W. This entire combination is connected across a supply of 100 V. Determine (i) effective resistance of the circuit (ii) total current.	L1 L1 L1 L3 L3	CO1	5 5 5 5
Section - 2				
3	a) Define (i) Capacitance (ii) Dielectric strength of a medium. b) Explain laws of electrostatics. c) Differentiate between a cell and a battery. d) Explain ampere-hour efficiency and watt-hour efficiency. e) Three capacitors have capacitances of 5μF, 8μF and 15μF respectively. Calculate the total capacitance when they are connected (i) in series (ii) in parallel.	L1 L2 L L2 L3	CO1 CO1 CO2 CO2 CO1	5 5 5 5 5
4	a) List the types and applications of inductors. b) Explain the effect of temperature on resistance and temperature co-efficient. c) List the applications of Ni-cd and Lead-acid batteries. d) Explain trickle charging of battery e) Three inductors of 60mH,120mH, and 75mH respectively are connected together in parallel combination with no mutual inductance between them. Calculate the total inductance of parallel combination.	L1 L2 L1 L2 L3	CO1 CO1 CO2 CO2 CO1	5 5 5 5 5

Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. **CIE Practice Test model question paper**

Program	**Electrical & Electronics Engineering**			**Semester**	**II**
Course Name	**Concepts of Electrical Circuits**			**Test**	**II**
Course Code	**24EE21IP**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					

Questions	**CO**	**Marks**
Note: Student has to Conduct the assigned question in Part-B and Part-A is common to all students. **Part-A** Identification of different types of capacitors and note down the specifications. Identification of different types of inductors.	**1**	**05**
Part-B 1. Verification of effective resistance by calculation and measurement a) Series Resistance Network b) Parallel Resistance Network /c) Series Parallel Network 2. calculate the specific resistance of Copper, Nichrome, (compare the values obtained and get the inference) 3. Verification of effective capacitance by calculation and measurement using LCR meter a) Capacitors in seriesb) Capacitors in Parallel/c) Capacitors in Series Parallel 4. Connect secondary cells in Series - Parallel group to a simple circuit and measure emf and current. 5. For a given circuit verify KCL. 6. For a given circuit verify KVL. 7. verify experimentally Thevenin's theorem. 8. Verify experimentally Maximum power transfer Theorem. 9. Demonstrate measurement of magnetic flux using flux meter. 10. Demonstrate the measurement of permeability of silicon steel.	**1,2**	**45**
Scheme of assessment		05

a)	Graded exercises	05
b)	Identification of passive components	10
c)	Circuit diagram, tabular column, procedure formulae, standard graphs if any.	20
d)	Conduction	05
e)	Results	05
f)	Viva-voce	
	TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	Report on different materials used for resistors, inductors and capacitors with their properties.
02	Report on applications of Thevenin's, Norton's and Maximum power transfer theorem/ Solve the given problems on Thevenin's/Norton's/Maximum power transfer theorems
03	Solve the given electrical network using KVL and KCL and submit the report.
04	Prepare a working model to demonstrate electromagnetic induction applicationsand submit the report.
05	Solve the problems on RL, RC and RLC series circuits to determine the total impedance, current, power and power factor. Draw the vector and wave diagrams.

9. Rubrics for Assessment of Activity(Qualitative Assessment)

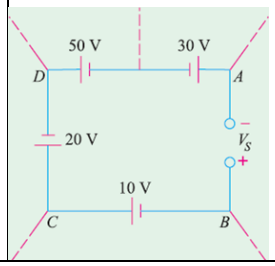
Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

10. SEE–Model Theory Question Paper

Program	**Electrical & Electronics Engineering**	**Semester**	**II**
Course Name	**Concepts of Electrical Circuits**	**Marks**	**50**
Course Code	**24EE21IT**	**Duration**	**90 Min**

Note: Answer any one full question from each section. Each full question carries equal marks.

Q No	Questions	Cognitive Levels	Course Outcomes	Marks
	Section -1			
1	a)Explain laws of electrostatics. b) Three capacitors have capacitances of 2μF, 3μF and 4μF respectively. Calculate the total capacitance when they are connected (i) in series (ii) in parallel.	L2 L3	CO1	5 5
2	a) Explain electro static induction and electric flux density. b) Twelve cells, each of emf 2 V and internal resistance 0.5 Ω, are connected in series across an external resistance of 4.5Ω. Determine (i) the current supplied by the battery (ii) terminal voltage of the battery.	L2 L3	CO1	5 5
	Section -2			
3	a) State and explain Kirchhoff's current law. c) Compare between primary cell and secondary cell with examples.	L2 L3	CO2	5 5
4	a)Explain Faraday's laws of Electrolysis. b)What is the voltage Vs across the open switch in the circuit of following figure using KVL. 	L2 L3	CO2	5 5
	Section -3			
5	a)a) Write the SI units of the following (i) flux (ii) mmf (iii) reluctance, (iv) flux density and (v) Absolute permeability.	L1	CO3	5 5

	b)d) A coil of resistance 100Ω is placed in a magnetic field of 1 mWb. The coil has 100 turns and a galvanometer of 400 W resistance is connected in series with it. Find the average e.m.f. and the current if the coil is moved in 1/10th second from the given field to a field of 0.2 mWb.	L3		
6	a) Define Magneto motive force (mmf), mention its unit and write the equation. b)Find the ampere turns required to produce a flux of 0.4 wb in a magnetic circuit of reluctance 100AT/wb.	L1 L3	CO3	5 5
7	a)Differentiate between the terms 'in- phase' and 'out of phase' alternating quantities. b)With a neat circuit diagram, wave diagram and vector diagram of R-L series circuit.	L2 L3	CO4	5 5
8	a)Define capacitive reactance, mention its unit and write its equation. b)Draw a sinusoidal waveform and mark the following (i) maximum value (ii) instantaneous value (iii) Time period (iv) frequency.	L2 L3	CO4	5 5
Section -5				
9	a)State and explain Lenz's law. b)Write the relationship between Voltage and Current in 3-ph Star system.	L2 L3	CO3 CO4	5 5
10	a)List the advantages of three phase supply over single phase supply b)Explain Effect of dielectrics on capacitance	L2 L3	CO3 CO4	5 5

11. **CIE- Model Practice Question Paper**

Program	**Electrical & Electronics Engineering**		**Semester**	**II**
Course Name	**Concepts of Electrical Circuits**	**Course Code - 25EE21I**	**Duration**	**180 min**
Questions			**CO**	**Marks**
Note: Student has to Conduct the assigned question in Part-B and Part-A is common to all students. **Part-A**			**1,2,23,4**	**10**

Identification of different types of capacitors and note down the specifications. Identification of different types of inductors. **Part-B** 1. Verification of effective resistance by calculation and measurement a) Series Resistance Network b) Parallel Resistance Network /c) Series Parallel Network 2. calculate the specific resistance of Copper, Nichrome, (compare the values obtained and get the inference) 3. Verification of effective capacitance by calculation and measurement using LCR meter a) Capacitors in series b) Capacitors in Parallel/c) Capacitors in Series Parallel 4. Connect secondary cells in Series - Parallel group to a simple circuit and measure emf and current. 5. For a given circuit verify KCL. 6. For a given circuit verify KVL. 7. verify experimentally Thevenin's theorem. 8. Verify experimentally Maximum power transfer Theorem. 9. Demonstrate measurement of magnetic flux using flux meter. 10. Demonstrate the measurement of permeability of silicon steel. 11. Rig up a circuit to demonstrate Faraday's laws of electromagnetic induction and the field due to a current carrying conductor. 12. Measurement of AC Voltage by CRO - Amplitude, Peak to Peak Value, Frequency, Time Period, Wavelength, and Phase Difference. 13. Measure power in a single phase AC circuit using ammeter, voltmeter, Wattmeter and calculate power, power factor for different loads like mercury vapour lamp/sodium vapour lamp/fluorescent lamp/LED lamp. 14. Measure experimentally the impedance of a series R-L circuit for different frequencies using function generator,CRO and verify it with theoretical impedance value. 15. Measure experimentally the impedance of a series R-C circuit for different frequencies using function generator,CRO and verify it with theoretical impedance value. 16. Measure experimentally the impedance of a series RLC circuit for different frequencies using function generator,CRO and verify it with theoretical impedance value. 17. Measure three phase power and energy in a three phase ac circuit.		**40**

18. Measure 3-ph. Star and delta voltages and currents. Verify with theoretical values.		
Scheme of assessment **a)** Graded exercises b) Identification of passive components c) Circuit diagram, tabular column, procedure formulae, standard graphs if any. d) Conduction e) Results f) Viva-voce		05 05 10 20 05 05
	Total Marks	**50**

1) Signature of the Examiner **2) Signature of the Examiner**

12. Equipment/softwarelistwithSpecificationforabatchof30students

Sl.No.	Particulars	Specification	Quantity
01	Regulated power supply	0-30V/60V,5/10A	10
02	Wire wound rheostats range-assorted	100 ohms, 1.5A	30
03	Portable DC Moving coil ammeter –multirange	(0-1-2-5-lOA)	30
04	Portable DC Moving coil Voltmeters –multirange	(0-2-10-15-30-75-150- 300V)	20
05	Portable Moving iron ammeter –multirange	(0-1-2-5-lOA)	20
06	Portable Moving iron Voltmeters -multirange	(0-5-10-15-30-75-150- 300-600V)	20
07	Multimeters analog and digital		06
08	Dynamometer type UPF Wattmeter	75/300/600V,5/10A	04
09	Knife switches SPST,DPST,DPDT,TPST,TPDT etc.		20
10	Galvanometers center zero type		10
11	Storage oscilloscope/CRO	100 MHz, Memory depth 500Mpts	05
12	Signal generator/Function generator	Sine, triangle, square	05
13	Thermometer industrial range		02
14	Standard screw gauge		02
15	Cells	1.5 v	20
16	Batteries	6v,12v,24v	05 each
17	Decade resistance boxes		05
18	Decade condenser boxes		05
19	Decade inductor boxes		05
20	Rheostats assorted		20
21	Dimmer stat/ single phase auto transformer	0-300 V, 10 A	05
22	Three phase auto transformer	0-600 V , 20 A	03
23	Resistive load/Lamp load	5 kW	05
24	Soldering iron	25W-10,60w-05	15
25	continuously variable power supply with current limit.	0-15 V at 2/lA	10
26	choke	230v,40Watts	05
27	bulbs	230v, 100 W	10
28	fan capacitor.	230v	05
29	continuously variable power supply with current limit	0-30V at 2/lA	10

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	ELECTRICAL AND ELECTRONICS ENGINEERING	**Semester**	**III**
Course Name	GENERATION, TRANSMISSION AND DISTRIBUTION	**Type of Course**	Integrated
Course Code	**25EE31I**	**Contact Hours**	8 hours/week 104 hours/semester
Teaching Scheme	L: T:P::4:0:4	**Credits**	6
CIE Marks	**50**	**SEE Marks**	**50 (Theory)**

1. Rationale:

The Generation, Transmission, and Distribution (GTD) subject is a cornerstone of electrical engineering, focusing on the journey of electricity from production to consumption. This subject is crucial for diploma students as it builds a comprehensive understanding of power systems, equipping them with the skills to address real-world energy challenges.

2. Course Outcomes: At the end of the Course, the student will be able to:

CO-01	Explain the significance of electrical power generation in modern society and analyze the economic factors involved in different generation methods, including the benefits of interconnected power systems.
CO-02	Identify key transmission line components and evaluate their parameters, including the performance of transmission lines under various conditions.
CO-03	Comprehend the design, function, and operation of substations, distribution systems, and transformer centers.
CO-04	Create and implement maintenance schedules for distribution transformers to ensure efficient, safe, and reliable operations within power distribution networks.

3. Course Content

WEEK	CO	PO	Theory	Practice
1	1	1,4,7	● Introduction-Importance of electrical power generation. ● Sources of energy available in nature.	video demonstration and prepare a report using Visit/Animations/ Video programme. 1 Types of hydro power plants, Classification of Hydro power plants https://youtu.be/2Ya1i7oQphg?si=PT3ykzf-PEyuP5fj 2. **Hydroelectric Power Plant working and components** https://youtu.be/_c9tBOjny28?si=uhltKzUW92AkB54t ● Locations of these different
	1	1,4,7	● Hydropower plant -Factors to be considered for selection of site ● Classify hydroelectric power plants based on the available head of water, plant capacity, load and construction..	
	1	1,4,7	● General layout of hydro power plant and explain its components. ● water hammering effect and its prevention.	

	1	1,4,7	• Advantages and Disadvantages of Hydroelectric power plant • Environmental Impact of Hydro power plants. • Micro-Hydro Power Plants (Types , Advantages,Challenges, Applications)	types of Large and Micro-Hydro Power Plants in Karnataka
2	1	1,4,7	• Thermal power plant- Factors to be considered for selection of site. • General layout of thermal (steam) power plant.	Sketch and interpret the schematic diagram of the thermal power station (T.P.S.) and its main cycles. https://youtu.be/WR1vDFLqdnc?si=OZooFilsk9JW4-O- Prepare technical report of visit to a nearby T.P.S.
	1	1,4,7	• Working of a thermal power plant. • Compare thermal power plant with hydroelectric power plant	
	1	1,4,7	• Advantages and disadvantages of Thermal power plants.	
	1	1,4,7	• Environmental Impact of Thermal power plants.	
3	1	1,4,7	• Nuclear power plant-Factors to be considered for selection of site . • Schematic diagram of nuclear power plant.	Sketch and interpret the schematic diagram of the nuclear power station (N.P.S.) https://youtu.be/edo1lozuwCs?si=gCpCX3oiugx1-4mo
	1	1,4,7	• Construction and working of Nuclear power plant • Nuclear power plant impacts on environment	
	1	1,4,7	• Nuclear waste disposal • environmental effect and precautions for nuclear power plants	
	1	1,4,7	• Diesel power plant-Schematic diagram of a Diesel generator unit and explain its components	
4	1	1,4,7	**Economics of Power Generation and Interconnected Power System** • Types of loads • Types and importance of load curve	Load curve of Campus/ Institute building(s) and calculation of following economical factors: Maximum demand, Average load, Load Factor, Reserve capacity, Plant capacity factor, utilization factor, Plant use factor and Diversity factor Prepare and interpret load curve for given data/data collected from nearby power station/substation
	1	1,4,7	• Terms and factors regarding load curve, Load duration curve	
	1	1,4,7	• Solve numerical related to load curve and load duration curve • Base load and peak load power plants	
	1	1,4,7	• Tariff - Meaning and Problems	
5	2	1,3,4,7	**Transmission Line Components, Parameters and Performance** • Explain AC transmission and distribution system with typical single line diagram, mention standard transmission and distribution voltages. (STRUCTURE OF POWER SYSTEM) • List the advantages and limitations of high voltage transmission.	• 1a. Identify Different components of Overhead Transmission lines. • 1b. Identify various conductors viz., All aluminium conductor (AAC, AAAC, ACSR conductor.) https://youtu.be/PaaUwwLR0v8?si=87JpSeCq0USgexJf
	2	1,3,4,7	• Classification of transmission lines based on distance and voltage level.	

			● Explain overhead transmission lines.	
	2	1,3,4,7	● Main components of overhead transmission lines. ● Explain the steps involved in erection of the transmission tower.	
	2	1,3,4,7	● Explain transmission through UG system ● Compare overhead and UG systems.	
6	2	1,3,4,7	● Classification of UG cables. ● list the types of UG cables based on construction. ● Explain with a diagram the general construction of a single core UG cable.	● Demonstrate various types of conductors used in overhead transmission lines. ● Video demonstration on construction of different types of HT/LT cables. ● Identify different parts of variousunderground cables. ● Select appropriate cable for a given application. https://youtu.be/aBI0InMxjGw?si=NzUT-1kcPRNt-KY5
	2	1,3,4,7	● Explain with a diagram the construction of 3 core XLPE cables. ● List and explain the methods of laying UG cable, ● mention their merits and demerits. List the faults in UG cables.	
	2	1,3,4,7	● explain line constants - resistance, inductance and capacitance. ● Short transmission line-equivalent circuit, vector diagram,	
	2	1,3,4,7	● write the equations for receiving end voltage, efficiency, voltage regulation and power factor.. ● Solve one simple example	
7	2	1,3,4,7	● Solve one simple example-Prepare the schedule of materials (only) for running of 50 KM 66 KV single circuit transmission line.	Simulation of Ferranti effect. https://youtu.be/YtYI9I8qAMU?si=Sx7TM_hlwBDHjFBl Solveproblems on Transmission Lines Determine voltage regulation and transmission efficiency of short transmission line. ● Practice preparation of cables for termination and joining. ● Demonstrate termination kits and practice on terminations of LT/HT cables.
	2	1,3,4,7	● Solve one simple example-Prepare the schedule of materials (only) for 110 KV & 220 KV single circuit transmission line.	
	2	1,3,4,7	● Explain Corona, formation of corona, factors affecting corona, advantages and disadvantages,	
	2	1,3,4,7	● Methods to reduce corona. ● Meaning of skin effect and Ferranti effect.	
8	2	1,3,4,7	HVDC, FACTS ● HVDC transmission lines-block diagram, ● list and explain the functions of main components of HVDC transmission system	HVDC Case Study 1,What are the primary advantages of HVDC transmission over traditional AC transmission, especially for long-distance power transfer? ● Focus on power losses, voltage regulation, and economic feasibility. 2, What are the different types of FACTS devices, and how does each
	2	1,3,4,7	● Types of HVDC links – monopolar, bipolar and Homo-polar DC links. ● Advantages Of HVDC transmission, Limitations of	

			HVDC transmission.	improve the performance of power transmission systems?
	2	1,3,4,7	● FACTSControllers- Definition, Objectives, ● Basic types of FACTS controllers and their functions.	
	2	1,3,4,7	● Compare HVDC and HVAC system	
9	3	1,3,4,7	SUBSTATIONS ● Meaning of substation and receiving station and their functions, ● Classification Of substations, ● Comparison between outdoor and indoor substation.	● **Prepare and submit a report by visiting nearby substation and observing the following** ● protection schemes ● Transformers used and their ratings ● earthing ● Bus bar and coupling arrangement ● Incoming lines and outgoing feeders ● Other components with their details
	3	1,3,4,7	● Code of practice related to substation erection. ● List the materials required for 66/11 KV substation with their specifications	
	3	1,3,4,7	● Bus bar arrangement-list the types-explain with diagram single bus arrangement with and without sectionalisation,	
	3	1,3,4,7	● Capacitor banks, specifications and calculation. selection of capacitor bank.	
10	3	1,3,4,7	● Explain the importance of interconnecting substations in large power systems. ● Functions of Load Dispatch Stations.	Read and Interpret Layout drawings of 220kV/110kV/66kV outdoor substations. Interpret various panel wiring drawings of substation equipment
	3	1,3,4,7	● Single Line Diagram (layout) of 33/11kV Sub-Station, 11kV/400V substation, symbols and functions of their components	
	3	1,3,4,7	DISTRIBUTION SYSTEM ● Classification of distribution systems. ● Explain with a diagram AC distribution system.	
	3	1,3,4,7	● List and explain the connection schemes of distribution system-radial, ring main and interconnected systems.	
11	3	1,3,4,7	● Distinguish between Feeder, distributor and service main.	Observe the various components of the Distribution System by visiting the MUSS and prepare a report on: 1. Number of feeders connected 2. Energy consumption of each feeder 3. Number of DTC meters connected 4. Percentage of distribution losses 5. 11kV Feeders Interruption Details
	3	1,3,4,7	● Concept of voltage drop in feeders/distributors with problem	
	3	1,3,4,7	● Prepare the schedule of materials(only) for 11 KV single circuit HT line for Rural Electrification ● Solve a problems on HT line	
	3	1,3,4,7	● 415 V LT lines-List theMaterials (only) with specifications and specify the method of calculating various LT line materials. ● Solve problems on LT line	

				6. Operation of 11kV feeders supplying power to IP sets in open delta
12	3	1,3,4,7	TRANSFORMER CENTRE ● Transformer center –Describe the types-Pole mounted,plinth mounted, indoor and outdoor types.	List the materials required for the 11 kV installation 1. Observe the substation erection and installation work 2. Observe the operation of distribution transformer 3. Check the poles set toproper depth, and are properly aligned 4. Observe the erection of channel on the pole 5. Observe the fixing of lightning arrester 6. Check the installationof earth connection as per standard procedure 7. Observe the lifting Observe the lifting of the transformer, to put it on the transformer bed in a safe and efficient manner 8. Observe the connection of low voltage cables 9. Identify Aerial bundled cables for LT and HT
	3	1,3,4,7	● Solve one simple example on determining the rating of Distribution Transformer. Write Specifications Of the Distribution Transformer.	
	3	1,3,4,7	● Prepare the schedule equipment/ Materials with specifications for a11KV/415V,100 KVA transformer center.	
	3	1,3,4,7	● Prepare the schedule equipment/ Materials with specifications for a11KV/415V,200 KVA transformer center.	
13	4	1,2,3,4,7	Maintenance schedule for distribution Transformer ● Explain the terms inspection, preventive maintenance and overhaul. ● Explain recommended schedule for inspection of Distribution transformers.	Visit the nearby substation and prepare a report on ● Maintenance schedule for distribution Transformer ● Maintenance schedule for 11kV overhead lines ● Maintenance schedule for 11kV UG system as per standard format
	4	1,2,3,4,7	Maintenance schedule for distribution Transformer ● Explain recommended schedule for preventive maintenance of Distribution transformers. ● Explain recommended schedule for overhaul of Distribution transformers.	
			Maintenance schedule for 11kV overhead lines ● Explain recommended schedule for inspection of 11kV overhead lines ● Explain recommended schedule for preventive maintenance of 11kV overhead lines.	

			Maintenance schedule for 11kV overhead lines ● Explain recommended schedule for overhaul of 11kV overhead lines	

4. References:

1. **Generation of Electrical Energy, by B.R.Gupta, publisher S.chand & company LTD, New Delhi**
2. **Electrical Power Generation, Transmission and Distribution. - S.N.Singh. PHI Publications.**
3. **Generation Distribution and Utilisation of electric energy by C.L. Wadwa, -New-Age International Publisher**
4. **Substation design and equipments-P.S Satna-Dhanpatrai.**
5. **Transmission, distribution and utilization – vol 3 B.L Theraja and A.K.Theraja.**
6. **Transmission and Distribution of Electric Power by J.B Gupta Katsons Publications.**

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2 Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4 Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

SEE - Theory Assessment Methodologies

Sl. No	SEE – Theory Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Theory	90	50	20

6. CIE Theory Test model question paper

Program	ELECTRICAL AND ELECTRONICS ENGINEERING			**Semester -III**	
Course Name	GENERATION,TRANSMISSION AND DISTRIBUTION			**Test**	**I/III**
Course Code	**25EE31I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	a) Explain why electrical power generation is important for daily life. b) What are the factors to be considered for selecting a site for a hydropower plant? c) Classify hydropower plants based on their capacity. d) Explain the general layout of a hydropower plant and its components	L3/L4	1	5+5+5+ 10-=25
2	a) What are the types of micro-hydro power plants b) What are the factors to be considered for selecting a site for a thermal power plant? c) Discuss the environmental impact of thermal power plants and ways to mitigate it d) Explain the general layout of a thermal power plant and its components.	L3/L4	1	5+5+5+ 10-=25
3	a) What are the factors to be considered for selecting a site for a nuclear power plant? b) Discuss the environmental impacts of nuclear power plants and ways to mitigate it c) Compare and contrast diesel generator units with other types of power plants. d) Explain the construction and working of a nuclear power plant.	L3/L4	1	5+5+5+ 10-=25
4	a) What is the meaning of tariff in electricity sector? b) Explain the importance of load classification in electrical power system. c) Explain the difference between a daily load curve and an hourly load curve. d) Discuss the impact of different types of loads on electrical power system design.	L3/L4	1	5+5+5+ 10-=25
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. CIE Practice Test model question paper

Program	ELECTRICAL AND ELECTRONICS ENGINEERING			**Semester**	**III**
Course Name	GENERATION,TRANSMISSION AND DISTRIBUTION			**Test**	**II/IV**
Course Code	**25EE31I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					

Questions	**CO**	**Marks**
1. prepare the report or PPT of Create a presentation comparing the types of hydro plants and their working principles. 2. Write a report identifying large and micro-hydro power plants in Karnataka with maps showing their locations 3. Create a detailed report or PPT explaining the working of a hydroelectric plant, with a focus on the major components. 4. Analyze how variations in water availability can impact plant performance 5. prepare the report or PPT : Sketch and Interpret the Schematic Diagram of a Thermal Power Station (T.P.S.) and Its Main Cycles 6. Prepare a Technical Report of a Visit to a Nearby T.P.S. 7. prepare the report or PPT : Sketch and Interpret the Schematic Diagram of a Nuclear Power Station (N.P.S.) 8. Prepare a technical report calculating and analyzing economic factors such as load factor, plant capacity factor, and utilization factor from collected data. 9. Present a load curve showing peak and base load patterns for a commercial or institutional building. 10. Identify the key components of overhead transmission lines, such as conductors, insulators, and towers. 11. Create a report or presentation on different types of conductors and their applications in overhead transmission systems. 12. Demonstrate the different types of HT/LT cables and explain their use. 13. Prepare a presentation on the termination and joining processes for underground cables. 14. Write a report on the preparation of cables for termination and joining for LT/HT cables. 15. Prepare a report on the voltage regulation and transmission efficiency analysis. 16. Demonstrate your understanding of the effect of line length on efficiency.		**50**
Scheme of assessment **a) Practical Demonstration-** ● Students will perform demonstrations related to the operation and components of hydro power plants, thermal power stations, and transmission lines, showcasing their understanding through hands-on activities and video presentations.		**15**
b) Technical Report- ● Students will prepare detailed reports based on site visits to power stations (thermal, hydro, nuclear) and analyze the working principles, components, and operational efficiency of the systems studied.		**15**
c) Presentation- ● Students will create and present PowerPoint presentations on specific topics, including types and classifications of power plants, load curves, and schematic diagrams of thermal and nuclear power stations.		**10**

d) **Problem-Solving Exercises**- ● Students will solve problems related to load calculations, voltage regulation, and transmission efficiency, demonstrating their analytical skills in applying theoretical knowledge to practical scenarios.	**5**
e) **Quizzes/Oral Exams** - ● Conduct quizzes or oral examinations covering fundamental concepts, key components, and operational principles of the studied power generation and transmission topics.	**5**
TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	- Visit to hydroelectric power plant, thermal power plants and D.G. power plant.
02	Study the distribution board, draw the (SLD) single line diagram of power distribution scheme in apartments/commercial buildings/hospitals/hotels. Show the details of rating of MCBs, fuses and switches.
03	Draw the distribution board /panel of your polytechnic. Draw the single line diagram of power distribution scheme and show the details of rating of MCBs, fuses and switches
04	Visit nearby 11kV, 66kV substations.
05	Identify parts of HT/LT cables, demonstrate termination kits, and practice cable joining.
06	**Retort**: Prepare a detailed case study report on HVDC transmission and the impact of FACTS devices on power transmission.

9. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	Students Score
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

10. SEE–Model Theory Question Paper

Program	ELECTRICAL AND ELECTRONICS ENGINEERING	**Semester**	**III**
Course Name	GENERATION, TRANSMISSION AND DISTRIBUTION	**Marks**	**50**
Course Code	**25EE31I**	**Duration**	**90 Min**

Note: Answer any one full question from each section. Each full question carries equal marks.

Q No	Questions	Cognitive Levels	Course Outcomes	Marks
	Section -1			
1	a) Explain the factors affecting the site selection for nuclear power plants b) With a neat Sketch diagram of hydro Electric power plant explain main components	L1/L2 L3/L4	1 1	3 7
2	a) Evaluate the nuclear waste disposal methods and their environmental implications. b) Design a schematic layout of a new thermal power plant, considering the major equipment and accessories	L1/L2 L3/L4	1 1	3 7
	Section -2			
3	a) Explain the importance of a load curve in power generation planning b) Compare the overhead and underground transmission systems.	L1/L2 L1/ L2	1 2	5 5
4	a) Describe the difference between base load and peak load power plants b) Explain the AC transmission and distribution system with a typical single-line diagram	L1/L2 L1/ L2	1 2	5 5
	Section -3			
5	a) Explain with a diagram the general construction of a single-core UG cable. b) Advantages Of HVDC transmission, Limitations of HVDC transmission	L1/L2 L1/ L2	2 2	5 5
6	a) Explain with a diagram the construction of 3 core XLPE cables.. b) HVDC transmission lines-block diagram	L1/L2 L1/ L2	2 2	5 5
	Section -4			
7	a) Prepare the schedule of materials for running a 50 KM 66 KV single-circuit transmission line	L3/L4	2	10
8	a) Create a step-by-step procedure for the erection of transmission towers for a 66 KV line.	L3/L4	2	10
	Section -5			
9	a) Single Line Diagram (layout) of 33/11kV Sub-Station, 11kV/400V substation, symbols and	L3/L4	3	5

	functions of their components b) Explain the recommended schedule for preventive maintenance of 11kV overhead lines	L3/L4	4	5
10	a) Prepare the schedule of materials(only) for11 KV single circuit HT line for Rural Electrification b) Explain the recommended schedule for the inspection of Distribution transformers	L3/L4 L3/L4	3 4	5 5

11. Equipment/software list with Specification for a batch of 30 students

Sl.No.	Particulars	Specification	Quantity
01	Computer Systems	smart class room	1
02	Projector and Display System	Miniature working model showing turbine, dam, and water flow	1
03	Hydro Plant Model	Miniature working model with boiler, turbine, and generator	1
04	Thermal Power Plant Model	Components: Reactor, steam generator, turbine	1
05	Nuclear Power Plant Model	Includes towers, conductors, and insulators	1
06	Electrical Transmission Line Model	Includes towers, conductors, and insulators	1
07	Cables and Accessories	HT Cables (11 kV and above) LT Cables (up to 1.1 kV) Termination kits and joining kits Cable cutters, strippers, and testers	
08	Tools for Wiring and Termination	Insulated screwdrivers Crimping tools Multimeters (Digital and Analog) Cable insulation testers (Megger)	

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	**Electrical & Electronics**	**Semester**	**III**
Course Name	Switchgear and Protection	**Type of Course**	Integrated
Course Code	**25EE32I**	**Contact Hours**	8Hrs/week 104Hrs/Sem
Teaching Scheme	L: T:P::4:0:4	**Credits**	6
CIE Marks	**50**	**SEE Marks**	**50 (Theory)**

1. **Rationale:** Switching and protective devices are vital for the control of various components such as generators, transmission lines, transformers, etc of a power system. Switchgear protects the system from the damage and ensures continuity of supply. Switchgear is an integral part of an electrical system in industries, hospitals, apartments, commercial and domestic buildings. An electrical and electronics diploma graduate should be capable to test, commission, trouble shoot and maintain electrical switchgear.

2. **Course Outcomes :** At the end of the Course, the student will be able to:

CO-01	Test and operate a given switchgear component.
CO-02	Choose the appropriate switchgear for a particular application and use them optimally.
CO-03	Test and provide protection schemes for power system components, such as generators, transformers, feeders.
CO-04	Test the substation grounding system and operate AMF, APF and STP.

3. **Course Content**

WEEK	**CO**	**PO**	**Theory**	**Practice**
1	**1**	**1,4**	Need for protective systems, nature and causes of faults, Definition of Symmetrical and unsymmetrical faults on three-phase systems, listing the types of unsymmetrical faults. Causes of Short-circuit, harmful effects of short-circuits,	Identification of different types of fuses and their applications. Measure and select the appropriate size of fuse wire.
			Definition of Switchgear, Differences between Indoor and Outdoor switchgear. components of a protection system, Listing of essential features of switchgear and commonly used switchgear equipments such as switches, fuses, circuit breakers and	

			relays. Zones of protection, primary and backup protection.	
			Fuse-protective device, advantages, disadvantages, composition. **Important Terms**: Current Rating of Fuse element, fusing current, fusing factor, Prospective current, cut off current, Pre-Arcing Time, Arcing Time, Breaking Capacity, Total Operating Time.	Test the HRC fuse by performing a Load test
			HRC fuses–list the types and applications, general construction and working, Merits and demerits.	
2	**1**	**1,2, 4**	**Circuit Breaker** – Meaning, Operating principle. Explain the terminologies– Arc Voltage, Restriking voltage, Recovery time, Arching Time, Pre–Arcing Time.	Test the MCB and plot its inverse time characteristic curve.
			Prospective Current, TRV, Recovery Voltage, RRRV, Total Break Time, Duties of circuit breaker under fault conditions, Making current, breaking current, Short circuit rating and Short-time current rating of circuit breakers.	Test any commercially available **ACCL, ELCB.**
			MCB-Types, applications, working, Comparison between fuse and MCB. Difference between fuse and circuit breaker.	
			Working principle and applications of ELCB, RCCB. Concept of **ACCL** (automatic changeover with current limiter).	
3	**1, 2**	**1, 2, 4**	Working principle and applications of MCCB, MPCB.	Test MCCB, MPCB.
			HV Circuit Breaker- OCB, advantages and disadvantages of oil as arc quenching medium, list the types of OCB, working principle of Plain oil circuit breaker, and disadvantages.	
			ACB, advantages, disadvantages, list types- explain, axial-blast and cross-blast types.	Visit to Substation. Identify the parts of circuit breaker and check its operation.
			SF6 Circuit Breaker- construction, working, advantages and disadvantages.	Video demonstration on Installation, operation and maintenance of OCB, ACB and SF6 circuit breaker.
4	**1, 2**	**1,4**	Problems on circuit breaker ratings.	Demonstration of various parts of an electromechanical relay.
			Protective Relay-Definition, explaining the typical relay circuit, Fundamental	Demonstration of time setting multiplier, pick up current setting for the relay operation.

			requirements of Protective Relaying. Classification of protective relays- based on technology, based on speed of operation, based on function.	
			Classification of protective systems. Importance of CTs and PTs for protection. Important Terms: pick up VA, Hold-on VA, drop out VA and Burden Pickup current, current setting, PSM, TSM, Time-PSM Curve. Problems.	Test Electromechanical overcurrent relay and plot the time vs current inverse characteristics of the relay.
			Construction, working principle and applications of Non-directional Induction type Electromechanical over current relays. Advantages and disadvantages of Electro-mechanical relays.	
5	**1, 2**	**1, 4**	Construction. Working principle and application of Earth Leakage relay. Meaning of ZCT (Zero-Phase Current Transformers), CBCT (Core balance CT)-construction, working and application.	1a) Rig up a circuit to test the earth leakage relay. 1b) Testing of Microprocessor based over current relay.
			Block diagram and working of Microprocessor/Microcontroller based over current relays.	
			Construction, working and applications of Static OV/UV and OCR relay- block diagram approach.	2a) Rig up the circuit to test static OV/UV relay 2b) Rig up the circuit to test Phase failure/phase reversal relay.
			Construction and operation of Numerical relay- block diagram approach.	
6	**1, 2**	**1, 4**	Advantages of Numerical relays, Comparison between Electromechanical, static and Numerical relays.	Program and test Numerical relay
			Causes of over voltages- Internal causes- switching surges, Insulation failure, Arcing ground, resonance and External causes-lightning, explain lightning phenomenon.	
			Lightning arresters, listing the types of lightning arresters, working principle of Rod gap and Horn gap arresters with advantages and limitations.	Video demonstration on functioning of lightning and surge arresters.
			Surge absorbers- definition and explanation of condenser type, choke and resistance type and Ferranti type surge absorbers.	
7	**3**	**1,4**	**Protection of transformers-** Common transformer faults, protection system for	1a) Test the operation of Buchholz Relay. 1b) Demonstrate the differential protection of a

			transformers, construction and operation of Buchholz relays, advantages and disadvantages of Buchholz relays.	transformer using Matlab Simulink.
			Transformer oil properties, applications of transformer oil.	
			Construction and working of Earth fault or leakage protection systems for transformer, Combined leakage and overload protection.	1a) Conduct BDVT on Transformer oil. 1b) A 3-phase transformer of 220/11,000 line volts is connected in star/delta. The protective transformers on 220 V side have a current ratio of 600/5. What should be the CT ratio on 11,000 V side?
			Construction and working of Circulating current scheme of transformer protection (Differential protection of transformer)	
8	**3**	**1, 4**	Protection of Alternators- Explain the abnormalities and list different types of faults in alternator.	Video demonstration on protection of alternator from faults.
			Differential protection of Alternators (Merz-Price circulating current scheme)- schematic arrangement, operation and limitations.	
			Stator-inter turn protection of alternators.	Simulate and study the waveform pattern for Alternator protection scheme.
			Construction and working of Balanced earth fault protection of alternators.	
9	**1, 3**	**1, 2, 4**	Feeder Protection- Explain the abnormalities and list different types of faults on feeders/ transmission line.	Simulate and study the waveform pattern for Feeder protection scheme.
			Time graded over current protection of a radial feeder using definite time relays and inverse time relays.	
			Parallel feeder, advantages of parallel feeders, Protection of parallel feeders.	Simulate/Test the operation of Differential Relay
			Differential pilot wire protection- Merz-Price voltage balance system.	
10	**1, 2, 3**	**1,2, 4**	Distance protection- principle and operation.	Simulation of Distance protection relay.
			Testing of Circuit breakers- Necessity of testing Circuit breakers, different testing methods of CBs.	
			Type tests on CBs- Mechanical test (Endurance test), Temperature rise test (Thermal test), Dielectric test.	Video demonstration on testing of circuit breakers
			Routine tests on CBs- list the types of routine tests, explain insulation resistance test, contact resistance test- millivolt	

			drop method.	
11	2, 3	4	Maintenance of circuit breakers.	1a) Video demonstration on testing of CTs 1b) Burden test on CT to determine its VA rating.
			Testing of CTs- list type and routine tests- Error measurement test- direct method, comparison method, turns ratio tests, exciting current test.	
			Polarity test, Insulation test, Over voltage inter turn tests.	Video demonstration on testing of PTs
			Testing of PTs- Type tests- Error measurement test, core insulation test, polarity test, list routine tests.	
12	3, 4	4	Substation Grounding- equipment grounding, system grounding, functions of earth mat.	Demonstrate Substation earthing.
			Neutral grounding, advantages of neutral grounding, Methods of neutral grounding- solid grounding, advantages and disadvantages of solid grounding.	
			Resistance grounding, advantages, disadvantages, reactance grounding, disadvantages, Peterson coil.	
			Bimetal relays/Thermal overload relays for motor protection- direct, CT operated	Testing of Motor protection against thermal over load.
13	4	4	EMR 5000 motor protection relay- key features, functions, benefits and applications.	Test the operation of APFC
			APFC- Automatic power factor control panel, working and applications.	
			AMF-Automatic mains failure detection panel, working.	Test the operation of AMF
			STP- Standard temperature and pressure control panel, working and application.	

References: (i) Principles of Power Systems by V.K.Mehta, Rohit Mehta

(ii) Power system protection and switchgear by Badri Ram, and D Vishwakarma

(iii) Sunil S.Rao, *"Switchgear Protection and Power Systems"*, 1999 11th edition, Khanna Publishers

(iv) J.B.Gupta "Switchgear & Protection", (edition), Katson Publisher,2008

(v) MadhavaRao T.S., 'Power System Protection - Static Relays', McGraw Hill, New Delhi,2nd Edition, 21st reprinted, 2008.

(vi) Testing , commissioning , operation and maintenance of electrical equipment by Sunil S Rao ,Khanna Publications

(vii) Power System Protection and Switchgear by Buvanesh A Oza, Nirmalkumar C Nair, Rases P Mehta and Vijay H Makwana, McGraw HILL Education (India Pvt. Ltd) New Delhi

(viii) https://www.youtube.com/watch?v=KwEdrgAoW0c

(ix) https://www.youtube.com/watch?v=vpcRY0SLt1o

(x) https://www.youtube.com/watch?v=Q9iEg0TRreE
(xi) https://www.youtube.com/watch?v=kEGTHhp1iV8
(xii)https://www.youtube.com/watch?v=T9_8xxp_TR8
(xiii) https://www.youtube.com/watch?v=iP4H9Tuu1Xo

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

6. SEE - Theory Assessment Methodologies

Sl. No	SEE – Theory Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Theory	90	50	20

7. CIE Theory Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester -III**	
Course Name	**Switchgear and Protection**			**Test**	**I/III**
Course Code	**25EE32I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					
Q.No	**Questions**		**Cognitive Level**	**Course Outcome**	**Marks**
Section - 1					

1	a) Define the term switchgear and list the essential features of switchgear. b) Explain the causes and effects of short-circuits. c) Explain operating principle of a circuit breaker. d) Explain the working of MCB. e) Compare fuse with MCB.	U Ap U Ap U	1 1 1 1 1	5 5 5 5 5
2	a) Define the term fuse and list the advantages of fuse. b) Explain the working of a HRC fuse. c) Discuss the terms (i) Arc Voltage and (ii) Restriking Voltage. d) Draw the circuit diagram of ACCL. e) Compare fuse with circuit breaker.	U Ap U Ap U	1 1 1 1 1	5 5 5 5 5
Section - 2				
3	a) Explain the construction and working principle of Sulphur Hexafluoride (SF_6) circuit breaker. b) Explain the term Protective Relay with the help of a typical relay circuit. c) List any three advantages and the disadvantages of oil circuit breakers.	Ap Ap Ap	2 2 2	10 9 6
4	a) A circuit breaker is rated as 1500 A, 1000 MVA, 33 kV, 3-second, 3-phase oil circuit breaker. Find (i) rated normal current (ii) breaking capacity (iii) rated symmetrical breaking current (iv) rated making current (v) short-time rating (vi) rated service voltage. b) Explain the principle of Earth leakage relay. c) List any three advantages and disadvantages of Air circuit breakers.	Ap Ap Ap	2 2 2	10 9 6
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. CIE Practice Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester**	**III**
Course Name	**Switchgear and Protection**			**Test**	**II/IV**
Course Code	**25EE32I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					
Questions				**CL**	**Marks**
1. Portfolio evaluation of practice sessions (week 1 to week 7)				**L4**	**05**
2. Demonstrate the operation of switchgear with suitable circuit and plot its characteristics (testing of a relay/fuse/MCB /MCCB/MPLCB)				**L4**	**40**
3. Viva-voce				**L4**	**05**

Scheme of assessment **a) Portfolio evaluation of practice sessions (week 1 to week 7)** **b)Writing the circuit diagram, procedure, standard graphs, etc.,** **c) Demonstration of testing of relay/fuse/MCB /MCCB/MPLCB) for a given operation.** **d) Results** **e)Viva-voce**	 **05** **15** **20** **05** **05**
Total Marks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Students and Faculty are encouraged to choose activities that are relevant to the topic. A group of 5 students can be formed to perform the activities. Each group should perform **two activities** and have to submit a report on them. The report has to be evaluated using suitable rubrics.

Sl.No.	Suggestive Activities for Tutorials
01	Prepare a hardware model consisting of PB switches (START & STOP) and contactor to run the three phase induction motor. Using earth leakage relay, trip the motor against earth faults and submit a report on this.
02	Prepare a hardware model consisting of PB switches (START & STOP) and contactor to run the three phase induction motor. Using Static OV/UV relay, trip the motor against voltage variations and submit the report on this.
03	Build Automatic mains failure model using contactors and relays and submit a report on this.
04	Build an Automatic power factor control model and submit the report on this.
05	Visit a manufacturing industry, study the protection circuits of the industry and submit a report on this.
06	Visit a substation, study protection mechanisms used by the substation and submit a report on this.

9. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

10. SEE–Model Theory Question Paper

Program	Electrical & Electronics Engineering	Semester	III
Course Name	Switchgear and Protection	Marks	50
Course Code	25EE32I	Duration	90 Min

Note: Answer any one full question from each section. Each full question carries equal marks.

Q No	Questions	Cognitive Levels	Course Outcomes	Marks
	Section -1			
1	a) Compare an outdoor switchgear with that of an indoor switchgear. b) Discuss the causes and effects of short circuit current.	L4 L4	CO1 CO1	5 5
2	a) Discuss the desirable characteristics of a fuse element with an example. b) A circuit breaker is rated 2500 A, 1500 MVA, 33 kV, 3 sec, 3-phase oil circuit breaker. Determine (i) therated normal current (ii) breaking current (iii) making current (iv) short time rating current.	L4 L4	CO1 CO1	5 5
	Section -2			
3	a) List the desired qualities/ features of protective relaying. b) With the help of a neat diagram explain the working principle of a core balance current transformer.	L2 L2	CO2 CO2	5 5
4	a) List any three advantages and two disadvantages of electromechanical type relays. b) Explain the causes of over voltages.	L2 L2	CO2 CO2	5 5
	Section -3			
5	a) Explain the construction and working of Horn gap type lightning arrestor. b) List any two advantages and one disadvantage of Buchholz relay.	L2 L2	CO3 CO2	7 3
6	a) Explain the working of earth fault or leakage protection system of transformer. b) List any three properties of transformer oil	L2 L1	CO3 CO2	7 3
	Section -4			
7	a) List any 4 abnormalities in alternators. b) Explain time graded over current protection of a radial feeder using definite time relays.	L1 L2	CO3 CO3	4 6

8	a) List different types of faults in feeders /transmission lines b) Explain the construction and working of balanced earth fault protection of alternator.	L1 L2	CO3 CO3	4 6
Section -5				
9	a) Explain mechanical endurance test on circuit breaker. b) Explain the phenomenon of Neutral grounding.	L2 L2	CO2 CO2	5 5
10	a) Explain error measurement test and polarity test on CTs. b) With the help of block diagram explain the working of AMF.	L2 L2	CO3 CO4	5 5

12. Equipment/software list with Specification for a batch of 30students

Sl.No.	Particulars	Specification	Quantity
01	Different types of fuses such as kit-kat fuse, cartridge fuse, glass fuse etc	----	2 Nos each
02	HRC Fuse with fuse base	10A	4 Nos
	MCB	Single pole, 2A	4Nos
03	Digital Time Interval Meter 0-999 ms, 0-99.9 sec, 0-99.9 min		4 Nos
04	ACCL- Automatic change over with current limiter	Single phase/ Three phase 6A/20A	4 Nos
05	ELCB /RCCB	2 pole, 32 A/ 220 V, 3 to 30 mA	4 Nos
06	MCCB- Moulded case circuit breaker	100A, 3Pole	4 Nos
07	MPCB-Motor protection circuit breaker	6-10 A, 3 pole	4 Nos
08	Transformer- Current injection	240/24 V, 20A 240/12 V, 50 A	4 Nos 1 No
09	Transformer- Voltage injection	240/600 V	4 Nos
10	Electromechanical relay - OCR/EFR / trainer kit	Current setting 0.5-10 A	1 Set
11	Earth leakage relay with CBCT	10 A or 16 A 415 V 30 mA or 100 mA ELR with 5 A/1A CBCT	2 Set
12	Static relay- OV/UV/Phase failure, reversal	415 V, OV setting of 105% to 120 % and UV setting of 95 % to 80%	2 Nos
13	Numerical relay/ trainer kit	Standard inverse curve 3s/10 or 1.3s/10 OR Very inverse/ Extremely inverse	1 set
14	BDVT test kit for testing the insulation strength of transformer oil.	220 V, 50 Hz 0-100 kV	1 set
15	Alternator protection module/panel	Protection against field failure, stator faults, frequency variations, prime mover failure etc.	1 set
16	Feeder protection module/panel	Time graded over current protection – definite/inverse time relays.	1 set
17	Current transformer	100/5 A, 50/5A	2 Nos.
18	Thermal over load relay	3 phase, 415 V, 0-4.5 A, or 0-6A, or 0-10 A	2 Nos.

19	Automatic mains failure kit/ panel	Single phase/ Three phase 6A/16 A 230 V /415 V	2 set
20	Automatic power factor control kit/ panel	440 V, 20 KVAr	2 set

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	**Electrical & Electronics Engineering**	**Semester**	**Third**
Course Name	**Generators & Transformers**	**Type of Course**	Integrated
Course Code	**25EE33I**	**Contact Hours**	7 Hrs/week 91 Hrs/sem
Teaching Scheme	L: T:P- 3:0:4	**Credits**	5
CIE Marks	**50**	**SEE Marks**	**50 (Practice)**

1. Rationale:

2. Course Outcomes : At the end of the Course, the student will be able to:

CO-01	Analyze the performance of DC Generators
CO-02	Analyze the performance of AC Generators
CO-03	Analyze the performance of Transformers
CO-04	Conduct maintenance on Generators and Transformers

3. Course Content

WEEK	**CO**	**PO**	**Theory**	**Practice**
1	**1**		Principle of working of DC Generator. Construction of DC Generator and functions of various parts.	Identification of various parts of DC Generator. Identification of Armature winding, Series field winding, Shunt field winding by Megger / Test lamp / Multimeter
	1		Derivation of EMF equation of DC Generator and related problems. Relationship between Electrical angle & Mechanical angle.	
	1		Classification of DC Generators. Circuit diagrams, Voltage & Current equations. Related problems	Connecting Separately excited, DC Shunt, Series DC Generators
2 **2**	**1**		Open Circuit characteristics & Load characteristics of DC Shunt, Series and Compound Generators.	

2	**1**		Power Flow stages in DC Generators. Various losses, Efficiency equation and Problems.	DC Shunt Generator Open circuit characteristics, Load characteristics - Internal and External
	2		Principle of working of 3-Phase AC Generator. Concept of Synchronous Generator. Construction of AC Generator and functions of various parts.	
3	**2**		Classification of Synchronous Generators, applications. Relationship between frequency, poles and Synchronous speed. AC windings, Types-Concentrated, Distributed, Full pitch, Factional Pitch Definitions and equations of Pitch factor, Distribution factor, Winding factor	DC Shunt Generator – Losses and efficiency by Direct Loading
3	**2**		Derivation of EMF equation of Synchronous Generator and related problems.	
3	**2**		Armature resistance, Synchronous Reactance and Synchronous Impedance. Open Circuit , Short circuit & Load characteristics of Synchronous Generators. Procedure, circuit diagram and equations. Equivalent circuit of Alternator	Identification of various parts of Alternator. Identification of Armature winding, field winding by Megger / Test lamp / Multimeter
4	**2**		Alternator on NO-Load and ON-Load. Vector diagrams On Load under Unity, lagging and Leading power factors. Armature reaction-Effect on terminal Voltage . Phasor diagram approach.	
4	**2**		Voltage Regulation and different methods of finding it. Equations for Voltage Regulation under Unity, lagging and Leading power factors. Related Problems.	OC test and SC test on Alternator Calculation of Armature resistance, Synchronous Reactance Synchronous Impedance
4	**2**		Parallel Operation of Alternators Necessity Conditions Circuit diagram – Two Alternators in Parallel, Alternator parallel to Infinite Bus bar	

5	**2**		Effect on Load Sharing between Alternators under ● Unequal Voltage ● Variation in Excitation ● Variation in Prime Mover Input **Static Excitation System**	**Calculation of** Pitch factor, Distribution factor, Winding factor
5	3		Transformer - Principle of operation, Construction,	
5	3		Functions of Main parts	**Load test on Alternator** **Estimate the Voltage regulation at** Unity, Lagging and Leading power factors.
6	3		Classification of Transformers	
6	3		Transformer cooling – Various methods, Properties of transformer Oil	Find the efficiency of alternator by conducting OC and SC test on it.
6	3		Emf equation of Transformer – Derivation, Turns Ratio, Transformation Ratio Related Problems	
7	3		Transformer on NO-Load, Phasor diagram, NO-Load losses, related problems	Connect two Alternators in parallel and share the load between them
7	3		Transformer ON Load, Resistance and Leakage Reactance, ON Load Losses Referred values, Equivalent Resistance and Reactance Equivalent circuit of Transformer referred to Primary side, Secondary side	
7	3		Phasor diagram of Transformer ON Load for Unity, Lagging and Leading power factors.	Connect one Alternators in parallel to grid share the load between them
8	3		Tests on Transformers Open circuit test, Short circuit test, Polarity Test Calculation of Equivalent circuit parameters from OC and SC tests	
8	3		Voltage Regulation in Transformers, Expressions for Voltage Regulation Unity, Lagging and Leading power factors. Related Problems	Identify different parts of Transformer, measure the primary & secondary winding resistance
8	3		Efficiency in Transformers Finding Efficiency from OC & SC tests Condition for Maximum efficiency - Derivation Related Problems	

9	3		All Day efficiency, related problems	Find the efficiency of 1-phase Transformer by direct loading
9	3		Three Phase Transformers – Working and construction Various Three Phase connections and its applications Star – Star Star – Delta Delta – Star Delta- Delta Open Delta (V-V) Scott (T – T)	
9	3		Vector Groups in Three Phase Transformers	Find the efficiency of 1-phase Transformer by conducting OC & SC test under different loads and power factors.
10	3		Parallel Operation of Transformers Necessity Conditions Advantages Two transformers in Parallel – Circuit connections	
10	3		Auto transformer	Find the voltage regulation of 1-phase Transformer by conducting OC & SC test under different loads and power factors.
10	3		Three winding Transformer	
11	3		Instrument Transformers Current Transformer Voltage transformer	Connect a Autotransformer to a load and apply variable voltage.
11	3		Phase Shifting Transformer, Furnace Transformer	
11	3		Welding transformer, High frequency Transformers	Connect two transformers in parallel and share the load.
12	3		Energy Efficient Transformers	
12	3		Transformers in HVDC Transmission	Measure the current by CT
12	4		Maintenance of Power transformers	
13	4		Maintenance of Distribution transformers	Demonstrate the maintenance of transformers
13	4		Maintenance of Alternators	
13	4		Overview of present Generation capacity in India	

4. References:

1) **Direct Current Machines – Rajput RK**
2) **Alternating Current Machines – Rajput RK**
3) **Electric Machines – Nagnath & Kothari**

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

6. SEE – Practice Assessment Methodologies

Sl. No	SEE – Practice Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Practice	180	50	20

7. CIE Theory Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester -3**	
Course Name	**Generators & Transformers**			**Test**	**I/III**
Course Code	**25EE33I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	**a)** Explain the functions of Yoke and Commutator in a DC Generator. **b)** Explain DC long shunt Cumulative and DC short shunt cumulative compound generators with circuit diagram, voltage and current equations. **c)** An eight pole lap wound DC generator has 960 conductors, a flux of 40 mWb per pole and is driven at 400 rpm. Find Open Circuit emf.	**5** **12** **8**		25
2	**a)** Explain the internal and external characteristics of DC Cumulative Compound generator.	**8**		

	b) Explain the power flow stages in a DC generator. **c)** A shunt generator supplies 195 A at 220 V. Armature resistance is 0.02 ohm, shunt field resistance is 44 ohm. If the iron and friction losses amount to 1600 watt, find (i) emf generated; (ii) copper losses; (iii) Efficiency	**7** **10**		
Section - 2				
3	**a)** Explain Concentrated, Distributed, Full pitch and Fractional Pitch windings. **b)** Derive the relationship between Synchronous speed, poles and frequency in a 3 phase Alternator. **c)** Determine the useful flux per pole required to develop 6600 V across the terminals of a 3-phase, star connected, 50 Hz, 4-pole alternator having 60 slots with 2 conductors per slot. The coils are short pitched such that if one coil side lies in slot No.1 and the other lies in slot No.13. Assume a double-layer winding	**10** **7** **8**		25
4	**a)** Draw the Phasor diagrams of Alternator on Load at Leading and Lagging power factors. **b)** Explain the conditions and connections of connection two 3 phase Alternators in parallel. **c)** A three-phase, star-connected, 1000 kVA, 3300 V, 50 Hz alternator produces a short-circuit current of 200 A and an open circuit voltage of 1040 V with the same field current 40 A. It its armature has an effective resistance of 0.2 ohm per phase, calculate full-load regulation of the alternator at upf, pf of 0.8 lagging and 0.8 leading.	**6** **10** **9**		
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. CIE Practice Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester**	**3**
Course Name	**Generators & Transformers**			**Test**	**II/IV**
Course Code	**25EE33I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					
Questions				**CO**	**Marks**

Conduct a suitable experiment to show that a DC Shunt Generator is a constant voltage generator. **OR** Conduct a suitable experiment to obtain the voltage regulation of a 3 phase Alternator at **a)** 75% of full load and 0.9 pf Lagging **b)** 50% of Full load and 0.6 pf Leading **c)** 100% of Full load and upf		**50**
Scheme of assessment **a)** Portfolio evaluation of experiments from Week-01 to Week-06 / Week-07 to Week-13 **b)** Circuit diagram with required components rating, specification and quantity. **c)** Step by step procedure of the experiment. **d)** Conduction of the experiment with suitable safety precautions and reading tabulation. **e**) Calculations / Graph plotting and final results f) Viva		**5** **10** **5** **10** **10** **10**
	TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	**For course outcome-01** Modern day sources and applications of Direct current electricity
02	**For course outcome-02** Modern excitation systems of Alternators.
03	**For course outcome-03** Various insulating materials and their properties in Transformers.
04	**For course outcome-04** Various maintenance on Transformers and Alternators

9. Rubrics for Assessment of Activity(QualitativeAssessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

11. **SEE- Model Practice Question Paper**

Program	**Electrical & Electronics Engineering**		**Semester**	
Course Name	**Generators & Transformers**	**Course Code 25EE33I**	**Duration**	**180 min**
Questions			**CO**	**Marks**
1) Conduct a suitable experiment to show that a DC Shunt Generator is a constant voltage generator. **2)** Conduct a suitable experiment to obtain the voltage regulation of a 3 phase Alternator at **a)** 75% of full load and 0.9 pf Lagging **b)** 50% of Full load and 0.6 pf Leading **c)** 100% of Full load and upf **3)** Conduct a suitable experiment to connect 3 Single phase transformers in three phase connections such that the line voltages at both primary and secondary terminals should not be same. **4)** Conduct a suitable experiment to obtain the voltage regulation and efficiency of a Single phase Transformer at **a)** 50% of full load and 0.6 pf Leading **d)** 80% of Full load and upf **e)** 100% of Full load and 0.9 pf Lagging **5)** Conduct a suitable experiment to connect 3 Single phase transformers in three phase connections such that the line voltages at both primary and secondary terminals should be same.				**50**
Scheme of assessment **a)** Portfolio evaluation of experiments from Week-01 to Week-13 **b)** Circuit diagram with required components rating, specification and quantity. **c)** Step by step procedure of the experiment. **d)** Conduction of the experiment with suitable safety precautions and reading tabulation. **e**) Calculations / Graph plotting and final results f) Viva				**5** **10** **5** **10** **10** **10**
			Total Marks	**50**

1) Signature of the Examiner **2) Signature of the Examiner**

12. Equipment/softwarelistwithSpecificationforabatchof30students

Sl.No.	Particulars	Specification	Quantity
01			
02			
03			

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	Electrical and Electronics Engineering	**Semester**	III
Course Name	Analog and Digital Electronics	**Type of Course**	Integrated
Course Code	25EE34I	**Contact Hours**	7 hours/week 91 hours/semester
Teaching Scheme	3:0:4	**Credits**	5
CIE Marks	**50**	**SEE Marks**	**50 (Practice)**

- **Rationale:**
  Analog Electronic circuits are used to amplify, process and filter analog signals using amplifiers, oscillators, switching circuits, operational amplifiers etc. Digital electronic circuits are usually made from large assemblies of logic gates. This digital logic circuitry is based on a binary system which has only two voltage levels, Low and High viz., Digital computers. Any intelligent electronic system is built by a combination of analog and digital circuits. Hence, it is imperative for any aspiring technician to acquaint with the concepts of analog and digital electronics.

- **Course Outcomes:** At the end of the Course, the student will be able to:

CO-01	Identify the components of a given electronic circuit, list the uses and characteristics of the components and test the components to ensure they are in working condition.
CO-02	Use datasheets to list the alternative electronic components for a given electronic circuit ensuring the results/output remains the same.
CO-03	Build an electronic circuit for a given application and demonstrate its working either in real or a simulated environment.
CO-04	Test a given circuit for desired result/outcome, identify the problem and troubleshoot to obtain the desired result/output.

- **Course Content**

WEEK	CO	PO	Theory	Practice
1	1	1,4	**ANALOG ELECTRONICS** 1.Passive components (Resistor, Inductors and capacitors): Introduction, symbols, units, types/classification, identification, selection and applications.	1. Identify different active and passive components. Mention their specifications.
			2.Semiconductors: Definition, types of semiconductor materials (Si, Ge, GaAs). -List characteristics of semiconductors by energy band diagram.	
			3. Intrinsic Semiconductors Doping a Semiconductor Extrinsic Semiconductors (N-type and P-type).	2a. Preparation of components, use of soldering iron and lead and flux. 2b. Standard Soldering practice to connect the components on

			-Concept of majority carrier and minority carriers. -Formation of PN junction and depletion region.	base-board/PCB/assembly board (follow soldering standard).
2	2,3	1,4	1.An overview of Junction Diode Zener diode- Symbol, Reverse bias characteristics	1.Test the working of Zener Diode as Voltage regulator.
			2.Zener Diode as Voltage regulator.	
			3.Bipolar Junction Transistors (BJT): Structure, Types, symbols, Construction, operation of (NPN/PNP) transistors.	2.Obtain output characteristics of BJT in CE configuration in physical mode. **OR** Simulate output characteristics of BJT in CE configuration.
3	1,3	1,4	1.BJT Configurations, Transistor current amplification factors- alpha, beta and mention the relationship between alpha and beta.	1a. Identify terminals of a MOSFET and test. b. Obtain output characteristics of MOSFET in physical mode. **OR** Simulate output characteristics of MOSFET.
			2. Input and output characteristics (CE Mode)- cut off, saturation, and active regions.	
			3.Transistor biasing- definition, importance, list types. -Explain Voltage divider bias. Transistor as a switch in CE mode. -Thermal runaway, necessity of heat sink. -Applications of transistor.	2.Construct and test MOSFET application circuit (MOSFET as a switch)
4	3,4	1,4	1. FET- Classification. MOSFET-Symbol, construction, working and characteristics, applications Compare BJT and MOSFET	1a. Identify the different types of fixed +ve and –ve regulator ICs and the different current ratings (78/79 series) 1b. Build +5V 1A DC Power Supply. **OR** Build +12V 1A DC Power Supply
			2.Rectifiers- half wave, full wave bridge rectifier- efficiency, ripple factor, PIV.	
			3.Filters- Necessity, Types- C and PI filters.	2a. Identify different heat sinks for IC based regulators b. Identify the parts, trace the connection and test the DC regulated power supply with safety. c. Troubleshoot and service a DC regulated power supply.
5	1,3	1,4	1.Regulated power supply- block diagram and applications. Regulator- 7805, 7809, 7812, 7905, 7912 Pin-Out details, operating voltages	1a. Identify the different types of LEDs , IR LEDs, Photodiode, photo transistor and list its applications. 1b. Construct a suitable circuit to demonstrate the working of photo diode.
			2.Amplifier- faithful amplification, classification based on configuration, power, and frequency.	
			3.Transistor RC coupled amplifier(CE Mode), Frequency response curve, band width	2.a. Construct and test a common emitter amplifier with bypass capacitors. 2.b. Construct and test a common emitter amplifier without bypass

			Working of Push pull amplifier.	capacitors.
6	1,3	1,4	1. OPAMP– block diagram, characteristics, µA 741 pin diagram.	1. Construct and test OP AMP as a Summer. Use an Analog IC tester to test the Analog ICs.
			2.OPAMP applications– inverting, non inverting amplifier, voltage follower.	
			3. Timer– block diagram, IC 555 pin diagram, duty cycle, time-delay, Applications, A stable multi-vibrator using IC 555.	2. Construct and test Astable timer circuit using IC 555. Construct and test mono stable timer circuit using IC 555. Use an Analog IC tester to test the Analog ICs
7	3	1,2	**DIGITAL ELECTRONICS** 1. Definitions- bit, nibble, byte, word, and parity bit. Number system- definition, types, radix, decimal, BCD, binary and hexadecimal.	1.Solve problems on number system
			2. Binary number system, Binary arithmetic: addition, subtraction, multiplication and division Complements: 1's, 2's.	
			3. Hexadecimal- addition, subtraction, Conversion- decimal to binary, decimal to hexadecimal	2. Solve problems on conversion.
8	3	1,2,4	1. Boolean variable, complement, Boolean function, expression, truth table. Boolean Algebra- rules and laws.	1. Realise basic gates using NAND gate.
			2.Logic gates NOT, AND, OR- definition, symbol, Boolean equation, truth table and working. Logic gates NAND, NOR, EX-OR-EX-NOR definition, symbol, Boolean equation, truth table and working. De Morgan's theorems- statement and equations	
			3. Karnaugh's map up to three variables- Simplification and drawing logic diagrams.	2. Realise basic gates using NOR gate.
9	1,3	1,4	1. COMBINATIONAL LOGIC CIRCUITS- Half adder- block diagram, logic diagram, truth table and working.	1a. Construct Half Adder circuit using ICs and verify the truth table 1b. Construct Full adder with two Half adder circuits using ICs and verify the truth table.
			2. Full adder- block diagram, logic diagram using truth table and working.	
			3. Multiplexer and Demultiplexer, 4:1 MUX, 1: 4 DEMUX List applications of MUX and DEMUX	2. Construct a circuit to verify the truth table of 4:1 multiplexer using IC 74153 and 1:4 Demultiplexer using IC 74139.
10	1,3	1,4	1. Encoders and Decoders- definition, applications. Seven segment display- working.	1.Construct a circuit to display 0-9 digits using standard Seven segment display with the help of decoder/ driver IC 7446/ or 7447.
			2. FLIP-FLOPS: S-R flip-flops, Clocked RS flip flop- block diagram, truth table, logic diagram.	
			3. JK flip-flop- block diagram, truth table, logic diagram.	2a. Identify different Flip-Flop ICs 2b. Verify the truth tables of Flip-Flop ICs (RS, JK) by connecting switches and LEDs.

11	1,3	1,4	1. Shift Registers- definition, types and applications. Four-bit SISO using D Flip flops- block diagram, truth table and operation.	1. Construct and test a four-bit SIPO register.
			2.Four-bit SIPO, PISO and PIPO shift registers using D flip flops- block diagram, truth table and operation.	
			3.Counters- definition, modulus concept, timing diagram, types and applications	2.Construct and test four-bit PIPO register.
12	3,4	1,4	1.Four-bit binary asynchronous counter- block diagram using JK flip flops, truth table, timing diagram and working	1.Construct and test 4-bit Asynchronous binary up /down counter
			2. Four-bit decade asynchronous counter- block diagram using JK flip flops, truth table, timing diagram and working.	
			3.Three-bit synchronous up counter- block diagram, truth table, timing diagram and working.	2. Rig up and test the truth table of Decade Asynchronous Counter
13	1,3	1,4	1.Digital to Analog converters: Binary weighted Resistor, DAC- block diagram and operation.	1.Construct and test (Binary weighted Resistor) Digital to Analog converter circuit.
			2. D/A converter specifications: resolution, accuracy and conversion speed. -Selection criteria for DAC	
			3. Analog to Digital converters: Successive Approximation ADC- block diagram and operation. -Selection criteria for ADC	2.Construct and test the Analog to Digital converter circuit.

4. References:

1. Electronic Devices and Circuits theory by Robert L. Boylestad Louis Nashelsky
2. Electronics Principles by Malvino, Mc. Graw Hill, Third edition. 2000.
3. Electronics Devices and Circuits by Allen Mottershead, PHI Learning Pvt. Ltd., First Edition
4. Electronics Principles and applications by Charles A Schuler and Roger L Tokhiem, Sixth Edition, Mc. Graw Hill, 2008. 2. 3.. 4.. 5.
5. Electronics Analog and Digital by I. J. Nagrath, PHI Learning Pvt. Ltd., 2013 Edition
6. Linear Integrated Circuits by Ramakant A. Gayakwad, PHI Learning Pvt. Ltd., Fourth Edition.
7. Work sheets digital 1. https://www.liveworksheets.com/un1107740kg 2. https://nohoacsl.weebly.com/uploads/1/1/2/2/112297219/ digital electronics worksheet.pdf

- **CIE Assessment Methodologies**

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
Total					50 **Marks**

- **SEE – Practice Assessment Methodologies**

Sl. No	SEE – Practice Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Practice	180	50	20

- **CIE Theory Test model question paper**

Program	Electrical and Electronics Engineering			**Semester -III**	
Course Name	Analog and Digital Electronics			**Test**	**I/III**
Course Code	25EE34I	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	a) Define Intrinsic and extrinsic semiconductor. Give example for each. b) Explain the working of PNP transistor with diagram. c) Explain how transistor is used as a switch in CE mode. d) Explain the operation of full wave bridge rectifier with the help of circuit and waveforms.	1 2 3 4	1 2,3 1,3 3,4	5 8 6 6
2	a) Define resistor and capacitor. Mention their symbol and units. b) Draw and explain the VI characteristics of a PN Junction Diode.	1 2	1 2,3	5 8

	c) Explain the concept of thermal runaway. How it is prevented? d) Explain the operation of half wave rectifier with the help of circuit and waveforms.	3 4	1,3 3,4	6 6
Section - 2				
3	a) List any 5 applications of transistor. b) Explain how Zener diode can be used as voltage regulator with the help of a circuit. c) Explain the characteristics of semiconductors using energy band diagram. d) Illustrate input and output characteristics of transistor in CE mode.	1 2 3 4	1,3 2,3 1 1,3	5 8 6 6
4	a) Define transistor biasing. List the different types of transistor biasing. b)Explain the reverse bias characteristics of a Zener diode with the help of a circuit and graph. c) Compare Intrinsic and Extrinsic semiconductors. d) Explain Voltage divider bias.	1 2 3 4	1,3 2,3 1 1,3	5 8 6 6
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

- **CIE Practice Test model question paper**

Program	Electrical and Electronics Engineering			**Semester**	**III**
Course Name	Analog and Digital Electronics			**Test**	**II**
Course Code	25EE34I	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					
Questions				**CO**	**Marks**
1.Identify terminals of a MOSFET and draw output characteristics of MOSFET. OR 2.Construct and test Astable timer circuit using IC 555.				1,3 1,3	**50**
Scheme of assessment **a)Portfolio evaluation of Practice Sessions** **b) Draw the Circuit diagram using the standard symbols with procedure and expected outcome.** **c) Build the circuit as per the circuit diagram** **d)Demonstrate the working of the circuit.** **e) Document the necessary readings of the circuit outcomes.** **f) Viva-Voce**					5 10 10 10 5 10
				TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

- **Suggestive Activities for Tutorials**:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	I. Study datasheets of diode 1N4001-1N4007, select a diode which has peak repetitive voltage (VRRM) of 100V, RMS Reverse Voltage of 70V, Average rectified current of 1 Amp and can withstand temperature of 150 degrees Celsius. Demonstrate the appropriate method of mounting the diode or its alternatives. II. Study given Handy Mobile Phone Charger Circuit, identify components required, test the components for working condition, build circuit in a bread board and test the circuit for desired output, if desired output is not obtained, troubleshoot to obtain desired result. Demonstrate the working of a circuit in the class. III. Demonstrate troubleshooting of simple DC circuit using any simulation software.
02	Study given basic LED Emergency Light with day light sensing Circuit, identify components required, test the components for working condition, build circuit in a bread board and test the circuit for desired output, if desired output is not obtained, troubleshoot to obtain desired result. Demonstrate the working of a circuit in the class. Demonstrate troubleshooting of LED Emergency Light circuit using any simulation software.
03	1.Build a (0 –30) V variable output regulated power supply using IC LM317 circuit. test the circuit for desired output and demonstrate the working of circuits in the class. 2. Demonstrate the output voltage of different IC 723 metal/ plastic type and IC 78540 regulators by varying the input voltage with fixed load.
04	Build an Automatic Fence Lighting Circuit with Alarm.
05	Build Adjustable Dual Timer Circuit using 555 Timer IC.
06	Two-Floor Elevator Numerous functions must be performed by the circuitry of an elevator (open/close door, move up/down, light up/down indicator, and so on). This example focuses on one aspect of a two-floor elevator: deciding when to move to the other floor. 1. Draw the truth table for given logical conditions. 2. write the logical expression. 3. Draw the logical diagram. 4. Implement the circuit. 5. Test the circuit for all possible input conditions.
07	A committee of three Individuals decides issues for an organization. Each individual votes YES or NO for each proposal that arises. A proposal is passed if it receives at least two YES votes. Design a circuit that determines whether a proposal is passed. 1. Draw the truth table for given logical conditions. 2.Prepare K-map and deduce simplified logical expression 3. Draw the logical diagram 4. Implement the circuit. 5. Test the circuit for all possible input conditions
08	Production line control Rods of varying length travel on conveyor belt ● Mechanical arm pushes rod within (+/- 5%) to one side ● Second arm pushes rods too long to other side ● rods that are too short stay on belt ● 3 light barriers (light source photocell) as sensors Design combinational circuit to activate arms. (Inputs are three sensors and outputs are two arm control signals)

09	Study 8-to-3 Bit Priority Encoder (74LS148). 1.Explain how it can be used in magnetic positional control as used on ships navigation or for robotic arm positioning etc 2. Draw truth table for above application 3. Draw block diagram and logic diagram

- **Rubrics for Assessment of Activity (Qualitative Assessment)**

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

- **SEE- Model Practice Question Paper**

Program	Electrical and Electronics Engineering		**Semester**	III
Course Name	Analog and Digital Electronics	**Course Code** 25EE34I	**Duration**	**180 min**
Questions			25EE34I	**Marks**
1. Test the working of Zener Diode as Voltage regulator. OR 2. Obtain output characteristics of BJT in CE configuration in physical mode. OR 3. Obtain output characteristics of MOSFET OR 4. Build +5V 1A DC Power Supply. OR 5. Construct and test a common emitter amplifier with/without bypass capacitors. OR 6. Construct and test OP AMP as a Summer. OR 7. Realise basic gates using NAND gate. OR 8. Realise basic gates using NOR gate. OR 9. Construct Half Adder/Full Adder circuit using ICs and verify the truth table OR 10. Construct a circuit to verify the truth table of 4:1 multiplexer using IC 74153 and 1:4 Demultiplexer using IC 74139. OR 11. Construct a circuit to display 0-9 digits using standard Seven segment display with the help of decoder/ driver IC 7446/ or 7447. OR 12. Identify terminals of a MOSFET and draw output characteristics of MOSFET. OR **13.** Construct and test Astable timer circuit using IC 555. OR			1 2,3 1,3 3,4 1,3 1,3 3 3 1,3 1,3 1,3 1,3	**50**

14. Verify the truth tables of Flip-Flop ICs (RS, JK) by connecting switches and LEDs. OR **15.** Construct and test four-bit SIPO/PIPO register. OR **16.** Construct and test 4-bit Asynchronous binary up /down counter OR **17.** Rig up and test the truth table of Decade Asynchronous Counter OR **18.** Construct and test the Analog to Digital converter circuit/ Digital to Analog Converter	1,3 1,3 1,3 3,4 3,4 1,3	
Scheme of assessment **a) Portfolio evaluation of Practice Sessions** **b) Draw the Circuit diagram using the standard symbols with procedure and expected outcome.** **c) Build the circuit as per the circuit diagram** **d)Demonstrate the working of the circuit.** **e) Document the necessary readings of the circuit outcomes.** **f) Viva-Voce**		 5 10 10 10 5 10
	Total Marks	**50**

- **Signature of the Examiner** **2) Signature of the Examiner**

- **Equipment/softwarelistwithSpecificationforabatchof30students**

Sl.No.	Particulars	Specification	Quantity
01	DC Regulated power supply	(0-30V, 2A)	10
02	DC Regulated Dual power supply	(+/- 15V,2A)	10
03	DC Regulated Dual power supply	(+/- 5, 1A	10
04	Cathode Ray Oscilloscope	Dual trace, 25 MHz	10
05	Signal Generator / Function generator	(5V P-P, 200mA)	06
06	DC Voltmeter	(0-1V)	10
07	DC Voltmeter	(0-10V)	10
08	DC Voltmeter	(0-30V)	10
09	DC Ammeter	(0 -100mA)	05
10	DC Ammeter	(0 -10mA)	10
11	Digital Multimeter	31 /2” 06	06
12	Analog Multimeter		06
13	LCR meter		02
14	Decade resistance box	(4 Dial)	10
15	Decade capacitor box	(4 Dial)	10
16	Analog IC Trainer Kit		10
17	Digital Trainer kit		10
18	Digital IC Tester		02
19	Electronic Circuit Simulation Software (Multisim/Pspice/ LT spice/GNU-Octave/ MatLabSimulink)		20

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	**Electrical & Electronics Engineering**	**Semester**	**Fourth**
Course Name	Electrical Motors & Drives	**Type of Course**	Integrated
Course Code	**25EE41I**	**Contact Hours**	8Hrs/week 104 Hrs/sem
Teaching Scheme	L: T:P-4:0:4	**Credits**	6
CIE Marks	**50**	**SEE Marks**	**50 (Theory)**

1. Rationale:

2. Course Outcomes : At the end of the Course, the student will be able to:

CO-01	Analyze the performance of Electric Motors
CO-02	Describe test parameters, testing procedures and demonstrate the troubleshooting of a given electric motor to ensure it performs optimally.
CO-03	Select a given motor for a specific application
CO-04	Construct power circuit and control circuits using appropriate components/devices to control the given electric motor.

3. Course Content

WEEK	**CO**	**PO**	**Theory**	**Practice**
1	1		Electrical Motors – Functions and general applications	Demonstrate the various industrial applications of Electric motors
1	1		Principle of working, general classification	
1	1		**DC Motor** – Construction Principle of working	Identify various parts of DC Motor
1	1		Concept of Back emf, Torque developed in DC Motor – Derivation Related Problems	
2	1		Classification of DC Motors Circuit diagram, Voltage and Current	Connect various DC motors

			equations of Series, Shunt and Compound DC Motors	
2	1,2		Electrical and Mechanical characteristics of DC Shunt, Series and compound motors	
2	1		Various losses, Power flow stages in DC Motor Applications of DC Motors	Obtain the Load characteristics of DC Shunt Motor
2	1		**Three phase Induction Motor** – Principle of Working Construction and classification	
3	1		Synchronous Speed, Concept of Slip, Slip Speed and Rotor frequency	Obtain the Load characteristics of DC compound Motor
3	1		Derivation of stator and rotor induced emf, Rotor current	
3	1		Power flow stages, various losses Relationship between Rotor input, Rotor copper loss and Rotor output Expression for Power output Related problems	Identify various parts of 3-phase Induction Motor
3	1,2		Equivalent circuit, Expression for Torque developed Torque-Slip Characteristics Condition for Maximum Torque Breakdown Torque / Pullout Torque Relationship between Starting Torque, Full load Torque and Maximum Torque Related problems,	
4	2		NO Load and Blocked Test on Three Phase Induction Motor Comparison between Induction Motor and Transformer Applications	Conduct load test on 3-phase Induction Motor and plot Torque-Speed characteristics
4	1,2		Induction Generators Operating principle Speed control – Slip power recovery scheme Applications	
4	1,2		**Synchronous Motor** Principle of Working Construction	Conduct No load and blocked test on 3-phase Induction Motor and find out the losses.
4	1		Expression for Power, Effect of load on Synchronous Motor	
5	1,2		Effect of Excitation on Armature current and Power factor V – Curves Inverted V-curves Synchronous Condenser	Identify various parts of 3-phase Synchronous Motor

5	1		Hunting Comparison between Induction motor and Synchronous motor Applications	
5	1,2		**Starting of Electrical Motors** Necessity of starting, Various methods employed	Start and run a 3-phase Synchronous Motor and plot a graph of Armature current with excitation.
5	1,2		**DC Motor** Starting – 3-point and 4-Point Starter	
6	1,24		**Induction Motor** Starting – Direct on line DOL Starting Star – Delta Starting Auto transformer starting Soft Starters	Start and run a DC shunt motor by 3-point and 4-point startter
6	1,2,4		**Synchronous motor** Starting Auxiliary Moor starting Starting as Induction Motor	
6	1,2,4		**Speed control of Electrical Motors** Need for speed control, Various methods of **acceleration** control Time Limit acceleration Current Limit acceleration Frequency Limit acceleration	Start, run a 3-phase Induction Motor by Manual / Automatic DOL Starter.
6	1,2,,4		**DC Motor** Speed control Armature voltage control Field Control Ward - Leonard	
7	1,2,4		**Induction Motor** Speed control Variation of Stator Voltage Rotor Resistance Control Pole changing method Variation of Supply Frequency	Start, run a 3-phase Induction Motor by Manual / Automatic Star Delta Starter.
7	1,2		**Synchronous motor** Speed control Pole changing method Variation of Supply Frequency	
7	1,2		**Protection of Electrical Motors** Limit Protection Incomplete sequence Protection Phase failure Protection Over voltage Protection Under voltage Protection and Under voltage release Protection Interlocking Protection Under frequency protection	Start, run a 3-phase Induction Motor by Manual / Automatic Autotransformer Starter.
7	1,2		**Braking of Electric Motors** Electric Braking – Different types, Necessity, Applications	

			Plugging, Dynamic Braking, Regenerative Braking Plugging in DC Shunt Motor, 3 Phase Induction Motor	
8	1,2		Dynamic Braking in DC Shunt Motor, 3 Phase Induction Motor Regenerative Braking in DC Shunt Motor, 3 Phase Induction Motor	Control the speed of DC shunt motor by armature voltage control
8	1,2		**Single Phase Motors** Working Principle, Construction and Applications Resistance Split, Capacitor start motor	
8	1,2		**Special Electric Motors** Working, construction and applications of Servo Motor, Stepper Motor	Control the speed of DC shunt motor by flux control
8	1,2		Working, construction and applications of PMDC Motor, Brushless DC Motor	
9	1,2		Working, construction and applications of Universal Motor, PM Synchronous Motor	Control the speed of a 3-phase Induction Motor by variation of Stator voltage.
9	1		**Industrial Drives** – Concept and Types Electric al Drives – Concept and Types, Classification, Block Diagram	
9	1		Quadrangle operation of Electrical drives, Constant Torque operation, Constant Power operation Constant Speed Drives, Variable Speed Drives, Variable Frequency Drive	Control the speed of a 3-phase Induction Motor by variation of rotor resistance.
9	3		**Selection of Motor power Rating** Classes of Motor Duty – Different Classes	
10	3		Problems on Motor power Rating	Control the speed of a 3-phase Induction Motor by variation of frequency.
10	3		**Selection of Electrical drives for Specific Applications** Factors to be considered for Selection of Motors	
10	3		Process and Drives selection for Steel Mills	Run and reverse the direction of DC Shunt motor by any one electric braking method.
10	3		Process and Drives selection for Cement Mills	
11	3		Process and Drives selection for Sugar Mills	Run and reverse the direction of 3-phase Induction Motor by any one electric braking method
11	3		Motors for Machine Tool Applications Cranes & Hoists, Compressors, Centrifugal Pumps, Mines	
11	1,2		**Traction Drives –**	U Start, Run and reverse the

			Classification of Electric Traction Types of Traction Services Current collection system	direction of 1-phase Induction Motor
11	1,2		Speed time curves Duty cycle of Traction Drives, Average speed, Scheduled speed Maximum speed	
12	1,2		Coefficient of Adhesion, Tractive Effort for acceleration Tractive Effort to overcome effect of gravity Tractive Effort to overcome train resistance	Connect and run a stepper motor, Servo Motor
12	1,2		DC Traction using DC Drive DC Traction using AC Drive AC Traction using AC Drive	
12	1,2		**Battery Powered Drives**	Calculate the required power rating of a motor for a given load condition.
12	1,2		**Solar Powered Drives**	
13	1		**Maintenance of Electrical Drives**	Demonstrate the operation of Traction drives for various types of current collection system.
13	1		Trouble shooting of Electric Motors	
13	1		**Energy conservation in Electrical Drives** Losses in Drive system Measures for Energy Conservation	Demonstrate the general maintenance of Electric Motors
13	1		Use of Energy efficient motors Use of Variable speed Drives Use of efficient Semiconductor converters Improvement of Supply Quality	Trouble shoot athe given faults in electric drive system

4. References:

1) **Fundamentals of Electrical Drives -Dubey GK**
2) **Electrical Drives concepts & Applications – Vedam Subramanyam**

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

SEE - Theory Assessment Methodologies

Sl. No	SEE – Theory Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Theory	90	50	20

6. CIE Theory Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester -4**	
Course Name	Electrical Motors & Drives			**Test**	**I/III**
Course Code	**25EE41I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	**a)** Explain DC seires, Shunt, Cummulative long shunt and Differential short shunt compound motors with voltage and current equations. **b)** Explain the power flow stages in a DC motor. **c)** The field resistance and armature resistance of a 120 V DC dhunt motor is 60 ohm and 0.2 ohm respectively. The motor draws 60 A current at full-load, 1800 rpm. If the brush contact drop is 3V, find the speed of the motor at half load.	**12** **6** **7**		

2	**a)** Explain the Torque – Speed characteristics of DC seires, shunt and differential compound motors. **b)** Derive the expression for torque in a DC Motor. **c)** A belt driven DC shunt generator runs at 1200 rpm delivering 10 kW at 220 V brushes. The belt breaks, following which the machine operates as a motor drawing 2 kW power. What will be its speed as a motor. The armature and field resistance are 0.25 : and 55:respectively. Ignore armature reaction and assume the contact drop at each brush to be 1V.	**9** **6** **10**		
Section - 2				
3	**a)**Explain the working of a 3-Phase Induction Motor. **b)Explain the speed control of an** 3-Phase Induction Generator by Slip power recovery scheme. **c)** A 3-phase induction motor draws 40 kW from the main, the corresponding stator losses are 15 kW. Calculate: (*a*) The total mechanical power developed and the rotor I^2R losses when the slip is 0 04. (b) The output power of the motor if the friction and windage losses are 0.8 kW. (c) The efficiency of the motor. Neglect the rotor iron losses.	**7** **10** **8**		
4	**a)**Explain the working of a 3-Phase Induction Generator. **b)**Give the power flow stages in a 3-Phase Induction Motor and obtain the relationship between rotor input, rotor output and rotor copper loss. **c) Compare Induction motor with Transformer.**	8 **10** **7**		
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

7. CIE Practice Test model question paper

Program	**Electrical & Electronics Engineering**			**Semester**	**4**
Course Name	Electrical Motors & Drives			**Test**	**II/IV**
Course Code	**25EE41I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					
Questions				**CO**	**Marks**
Conduct a suitable experiment to show that a DC Shunt motor is a constant speed motor.					**50**

OR Conduct a suitable experiment to start a 3 phase Induction motor such that only 57,7% of voltage is applied to its phase windings at the time of starting.		
Scheme of assessment **a)** Portfolio evaluation of practice sessions from Week-01 to Week-06 / Week-07 to Week-13 **b)** Circuit diagram with required components rating, specification and quantity. **c)** Step by step procedure of the experiment. **d)** Conduction of the experiment with suitable safety precautions and reading tabulation. **e**) Calculations / Graph plotting and final results f) Viva		**5** **10** **5** **10** **10** **10**
	TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

8. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	**For course outcome-01** Various electric traction services provided by Indian Railways
02	**For course outcome-02** Modern speed control methods for Induction motor
03	**For course outcome-03** Present status of AC and DC drives
04	**For course outcome-04** Energy conservation measure used in a given Industry
05	
06	

9. Rubrics for Assessment of Activity(QualitativeAssessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

10. SEE–Model Theory Question Paper

Program	**Electrical & Electronics Engineering**	**Semester**	**4**
Course Name	Electrical Motors & Drives	**Marks**	**50**
Course Code	**25EE41I**	**Duration**	**90 Min**

Note: Answer any one full question from each section. Each full question carries equal marks.

Q No	Questions	Cognitive Levels	Course Outcomes	Marks
	Section -1			
	a) Explain any two current collection systems in Traction drives. **b)** Draw the equivalent circuit of a 3 Phase induction motor reffered to rotor side.	A	1	5 5
	Section -2			
	a) Explain the V-curves and Inverted V-curves in a Three phase Synchronous motor. **b)** A 400 V, 6-pole, 50 Hz, 3-phase induction motor develops 20 H.P. inclusive of mechanical losses when running at 995 rpm, the power factor being 0.87. Calculate: (i) Slip (ii) The rotor copper losses (iii) The line current. The stator copper loss is 1500 W.	A	1	5 5
	a) Explain the speed torque characteristics of a 3 phase induction motor for various values of rotor resistance. **b)** A four-pole DC shunt motor draws a current of 36 A when connected to 230 V mains and delivering 10 hp. If armature and field circuit resistances are 0.1 : and 300 : respectively, find the motor current when load is 5 hp. Neglect armature reaction effect.	A	1	5 5
	Section -3			
	a) Give the applications of Servo motor and BLDC motor. **b)**Explain any one energy conservation measure for minimising losses in a electric drive.	A	1,4	5 5
	a) Give the applications of Universal motor and PM Synchronous motor. **b)** Give the troubleshooting steps when an 3-Phase AC motor fails to start.	A	1,4	5 5

Section -4				
	a) Explain any one reduced voltage starting method for starting a 3 Phase induction motor. **b)** Explain the Under voltage protection and Under voltage release protection.	A	2	6 4
	a) Explain the Regenerative braking in DC Traction employing DC Drive. **b)**Explain the speed control of a AC motor by variation in frquency,	A	2	6 4
Section -5				
	a) Explain the main functions of various blocks in an electric drive. **b)** Select suitable motors for Kiln drive and Crusher drive in cement plant with justifications.	A	3	5 5
	a)Explain Intermittent periodic loading and Coutinuos loading motor duties. **b)** Select suitable motors for Hoists and compressors with justifications.	A	3	5 5

12. Equipment/softwarelistwithSpecificationforabatchof30students

Sl.No.	Particulars	Specification	Quantity
01			
02			
03			

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	**Power Electronics**	**Semester**	**IV**
Course Name	ELECTRICAL & ELECTRONICS ENGINEERING	**Type of Course**	Integrated
Course Code	**25EE42T**	**Contact Hours**	8Hrs/week 104Hrs/Sem
Teaching Scheme	L: T:P 4:0:4	**Credits**	6
CIE Marks	**50**	**SEE Marks**	**50 (Theory)**

1. Rationale:

Power electronics is a subject that concerns the applications of electronic principles in situation that are rated at power level rather than signal level. Power electronics is the application of solid-state electronics to the control and conversion of electric power. Power semiconductor devices are used to construct converters and inverters in the various applications such as power supplies, Electric drives, Flexible AC Transmission systems and Distribution systems, EV's, Energy storage devices. An Electrical Technician shall apply the knowledge of Power electronics to control and convert Electrical Power for an application.

2. Course Outcomes: At the end of the Course, the student will be able to:

CO-01	Identify the power electronic devices using relevant datasheets and demonstrate its suitability to produce specified electrical and thermal characteristics.
CO-02	Build a power electronic circuit for a given application, demonstrate its working either in real or simulated environment.
CO-03	Test a given power electronic circuit, identify the problem and troubleshoot to obtain the desired result/output.

3. Course Content

WEEK	**CO**	**PO**	**Theory**	**Practice**
1	1,2	1,4	What is Power Electronics and Why it is required?Evolution of Power Electronics.Basic Components of Power Electronics.(General Block Diagram of Power Electronic Circuit).Advantages and disadvantages of Power Electronics.	1 a)Study and interpret the datasheet of the given power diode and BJT. 1b. Test the given Power Diode and BJT for its working condition.
	1,2	1,4	Draw the layer structure of the power diode and explain V –I characteristics of power diodes. Types of power diodes and their applications	

		1,4	Types of Power transistors, BJT input and output characteristics. Operation of BJT as a switch.	2a) Rig up a circuit to plot the VI characteristics of the power diode. 2.b) Rig up a circuit to operate BJT as a switch.
2	1,2	1,4	1. Operation of N-channel enhancement power MOSFET, and its transfer characteristic curve, Application of MOSFET.	Study and interpret the datasheet of the given IGBT andMOSFET (H20R1203 IGBT power transistor - 20A / 1200V , **IGBT 25N120R FSC 25N120 – 25A 1200V IGBT Transistor** **IRFP460 MOSFET - 500V 20A N-)** 2a. Rig up a circuit to plot transfer and output characteristics of power MOSFET and IGBT. 2b. Rig up the circuit of power MOSFET as a switch (https://circuitdigest.com/electronic-circuits/simple-mosfet-switching-circuit-how-to-turn-on-turn-off-mosfets)
	1,2	1,4	2. Structure of IGBT and its characteristics	
	1,2	1,4	3. Operation of MOSFET as switch. 4.Application of IGBT, Compare MOSFET, BJT and IGBT.	
3			Necessity of driver circuit for MOSFET, types of MOSFET driver circuits- Optocoupler isolated, transformer isolated, non-isolated driver circuits.	https://microcontrollerslab.com/mosfet-driver-ir2110-pinout-examples-applications-datasheet/ Test a given MOSFET driver IC(**IR2110 as a high side MOSFET driver** **IR2110 as a low side MOSFET driver)**)
			List driver ICs for MOSFET. **Pin diagram and connection diagram of MOSFET IC DRIVER (** IRS44273L) Steps to select a power MOSFET- Select a suitable MOSFET for 12V INVERTER for a residential application. **https://www.electronicsforu.com/buyers-guides/guide-to-select-power-mosfets** Select a suitable MOSFET for a given application. (https://www.microtype.io/selecting-a-mosfet-a-pragmatic-approach/)	
			Necessity of driver circuit for IGBT	https://www.youtube.com/watch?v=eQ-QDtw4R9g Test a given a IGBT driver IC .
			List driver ICs for IGBT. IGBT selection criteria- select suitable IGBT for a given application.	
4	1,2	1,4	1.Layer diagram of SCR and Concept of two transistor analogy of SCR.	1.Test the given SCR for its working condition. 2. Rig up the circuit to plot VI characteristics of SCR.
	1,2	1,4	2. Static V-I characteristic curve of SCR, Enumerate Reverse blocking, Forward blocking, forward conduction mode. List different types	

			of SCR.	
	1,2	1,4	3. GTO, principle of operation and list its application,	
			4.Layer structure of LASCR and explain its operation.	
5	1,2	1,4	1.Layer structure, operation and characteristics of TRIAC	1a. Test the given TRIAC and DIAC for its working condition. 1b. Simulate VI characteristics of TRIAC and DIAC. 2.Build and test a TRIAC- fan motor speed control circuit.
	1,2	1,4	2. 4-Modes of turn on of TRIAC and state the preferred mode of turn-on.	
	1,2	1,4	3. Operation of DIAC and its V-I characteristic, curve, application of DIAC.	
			4.**SCR Control Circuits:** Methods of turn on of SCR	
6	1,2	3,4	1.General layout of firing circuit.	1a..Build R firing circuit and determine the maximum firing angle. 1b.. Build R – C firing circuit and determine the maximum firing angle. 2a..Build and test UJT Relaxation oscillator. 2b.. Build and Test time delay relay using SCR and UJT.
	1,2	3,4	2. Operation of R firing circuit and R-C firing circuit with waveforms.	
	1,2	3,4	3. Construction, operation and characteristics of UJT	
			Digital firing scheme with waveforms Voltage and current ratings of SCR and Reliability of SCR, MTBF	
7	1,2		**MOSFET protection circuit**s- over current , over voltage and short circuit protection.	Build a MOSFET protection circuit for a given MOSFET. Build a protection circuit for a given IGBT Identify different IGBT modules (https://www.infineon.com/cms/en/product/power/igbt/igbt-modules/)
	1,2		IGBT protection circuits- over current, over voltage protection. IGBT module: Electrical feature and application	
	1,2		How SCR can be protected against overvoltage and over current, di/dt & dv/dt.	
	1,2		Thermal protection- necessity of heat sinks for MOSFET, IGBT, BJT, SCR. Selection of heat sink. Mounting of MOSFET, IGBT, BJT, SCR on heat sinks	Construct a snubber circuit for protecting SCR use freewheeling diodes to reduce back emf.
8	1,2	3,4	.Commutation:- line commutation, forced commutation and methods of forced commutation	Simulate Line commutation and forced commutation circuits and observe waveforms. 1.Build single-phase full controlled bridge converter for resistive load. Trace the waveforms across SCR and load. 2. Simulate single-phase Dual converter and observe waveforms 3.Test and operate Active Bridge Rectifier Controller(TEA2209T) https://www.nxp.com/products/power-management/ac-dc-solutions/ac-dc-controllers/active-bridge-rectifier-controller:TEA2209T
	1,2	3,4	Meaning of Load commutation and complementary commuation.	
	1,2	3,4	**CONVERTERS:** types of power electronic converters: Single -quadrant semi-converter, two- quadrant full- converter and dual converter.	

9	1,2	3,4	1. Single-phase full converter RLE type with continuous load current 2.single-phase dual converter and principle of operation.	
	1,2	3,4	3.Gating pulse requirement of three-phase full converters.	. Test and operate three-phase bridge rectifier module(36MT160) .
	1,2	3,4	4.Three-phase bridge converter and 180 conduction mode with waveforms	
10	1,2	4	1.Inverter- Types of inverters (VSI and CSI) 2.Half bridge and full-bridge inverter and its operation 3.Introduction to PWM inverters and list the types of PWM techniques. 4.Three-phase inverters.	.Simulate Full-bridge Inverter and observe waveforms. Build 2000w sinewaveinverterusing IGBT andMOSFET https://www.youtube.com/watch?v=cNxYYwIyT0A **(How to make a Powerful Inverter 12v DC to 220v AC using TL494(PWM) 50hz 60hz Invereter https://www.youtube.com/watch?v=9KRXIZY3RSQ)** **(Introduction to Single Phase Inverter INV 01 using Arduino - Part 01** **https://www.youtube.com/watch?v=uPZZP_oMpjM)** 2.Build and test an single phase inverter.
11	2,3 2,3	4	1.. DC Chopper- Step-down and Step-up chopper and its operation 2.. Different chopper configurations- (A, B, C, D and E). 3. Cycloconverter- Draw the circuit of mid- point Step-down cyclo-converter and its operation with waveforms 4. Cycloconverter- Draw the circuit of a mid- point step-up cycloconverter and its operation with waveforms	Simulate/Build step-down chopper and observe waveforms. 2. Simulate / Build step-up chopper and observe waveform 1.MOSFET BASED PWM CHOPPER / PWM BASED STEP DOWN DC CHOPPER USING MOSFET
12	2,3	4	**DC-DC converters:** 1. Working principle of Buck converter (regulator). 2. Working principle of Boost	1 1a. Build & test IC based - DC –

			converter (regulator). **Ref.7(9,10,11,12)** 3. Working principle of Buck-boost and flyback converter. **Ref.7(14,15,16,17)** 4. SMPS and its operation and application.	DC converter for different voltages. **Ref.7(8)** 1b. Test the monolithic synchronous buck regulator (MP2305). 2.Test and Troubleshoot regulators. **Ref.7(13)**
13	2,3	4	1. Draw the block diagram of offline online UPS and its operation. 3b. List basic troubleshooting steps for UPS. **Ref.7(21,23)** 2.Sizing of UPS and Battery for a given load and load power factor . 3. Power quality: power line disturbances- over voltage, black out, voltage spike, chopped voltage waveform, voltage fluctuations. 4. Harmonics: Source of harmonics , THD, distortion factor. frequency distortions,	1a. Identify various input and output sockets/ connectors of the given SMPS. 1b. Identify major sections/ ICs/components of SMPS. 1c. Troubleshoot given SMPS. **Ref.7(19,20)** 2a. Identify front panel control & indicators of UPS. 2b. Connect battery & load to UPS & test on battery mode. 2c. Open top cover of UPS & identify isolator transformer & UPS transformer & additional circuit other than inverter. 2d. Identify various circuit boards in UPS and monitor voltages at various test points. 2e. Test UPS under fault condition & rectify fault. **Ref: 7(22,24)** 2e. Calculate size of UPS and battery size for give load https://www.luminousindia.com/load-calculator https://www.india.fujielectric.com/resources/technical-guide/ups-sizing-calculation

4. References:

Sl. No	Description
1	Power Electronics by Dr PS Bimbhra, Khanna Publishers, New Delhi
2	IndustrialElectronicsandControlBiswanathPaulPHIPublicationEdition-II
3	Thyristorised power controllers GK Dubey
4	Power and industrial Electronics by Harish C Rai
5	Power electronics by Mohan Undeland Robbins, Wiley Publications
6	Modern Power Electronics by PC. Sen
7	Power Electronics–Raghunath Rao

8	https://components101.com/ics/ncp3064-dc-dc-converter-ic
9	https://www.youtube.com/watch?v=4IQBN6Oy8Lg
10	https://www.youtube.com/watch?v=Rf29oUGpwI
11	https://www.youtube.com/watch?v=vmNpsofY4-U
12	https://www.youtube.com/watch?v=yD7fMylYgXw
13	https://www.youtube.com/watch?v=-3tBw6WSZVM
14	https://www.youtube.com/watch?v=zNfbbPobtus
15	https://www.youtube.com/watch?v=ZiDX-uoTQ
16	https://www.youtube.com/watch?v=9--jaxiXhE
17	https://www.youtube.com/watch?v=Fk-B6OO6GB8https
18	https://www.youtube.com/watch?v=6CsTIPjFZ48www.youtube.com/watch?v=I8c5DLJgS3o
19	https://www.youtube.com/watch?v=PPLjXkca7eo
20	https://www.powersupplyrepairguide.com/previewsmpsebook.pdf
21	https://www.youtube.com/watch?v=C1BYo88HSU0
22	https://www.youtube.com/watch?v=3oq18dZmb3Q
23	https://www.apc.com/lr/en/faqs/FA279110/
24	https://www.youtube.com/watch?v=lkriUIUdflM https://www.youtube.com/watch?v=JHgKBDoQCyQ
25	https://inst.eecs.berkeley.edu/~ee100/su07/handouts/EE100-MultiSim-Tutorial.pdf
26	http://eceweb1.rutgers.edu/~psannuti/ece223/Manual-for-multisim.pdf
27	https://www.multisim.com/help/getting-started/

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	

5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

6. SEE - Theory Assessment Methodologies

Sl. No	SEE – Theory Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Theory	90	50	20

7. CIE Theory Test model question paper

Program	**ELECTRICAL & ELECTRONICS ENGINEERING**			**Semester -IV**	
Course Name	**POWER ELECTRONICS**			**Test**	**I/III**
Course Code	**25EE42I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					
Q.No	**Questions**		**Cognitive Level**	**Course Outcome**	**Marks**
	Section - 1				
1	a)List advantages of power electronics system b)Explain the operation of N-channel enhancement power MOSFET, c) Which power transistor is most suitable for Inverter application? Justify your answer. d) Explain the input and out put characteristics of BJT		L1 L2 L3		5 5 5 10
2	a) List the applications of MOSFET b)Explain the structure of IGBT c)Compare MOSFET, BJT and IGBT d)Explain the transfer characteristics of IGBT and MOSFET.				5 5 5 10
	Section - 2				
3	a)List the types of Driver Circuits for MOSFET b) Explain the need for the driver circuit for IGBT c) Explain the steps to select a power MOSFET- for 12V		L1 L2 L3		5 5 10

	INVERTER for a residential application. d) Explain the operation of GTO with a neat sketch.			5
4	a) List different types of SCR b)Explain the need for the driver circuit for MOSFET c) select suitable power transistors for the following with proper justifications i.) High frequency applications ii) High power AC applications d) Explain the operation of LASCR with neat sketch	L1 L2 L3		5 5 10 5
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

8. CIE Practice Test model question paper

Program	**ELECTRICAL & ELECTRONICS ENGINEERING**			**Semester**	**IV**
Course Name	**POWER ELECTRONICS**			**Test**	**II/IV**
Course Code	**25EE42I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					
Questions				**CO**	**Marks**
Rig up a circuit of to operate power MOSFET as a switch				**2**	**50**
Scheme of assessment **a)** Portfolio evaluation of practice sessions from Week-01 to Week-06 / Week-07 to Week-13 5 **b) Demonstrate the working condition of power electronics devices (MOSFET). 5** **c) Drawing of the Circuit diagram using the right symbols 5** **d) Build the circuit as per the circuit diagram 10** **e) Demonstrate the working of the circuit 15** **f) Document the necessary readings of the expected circuit outcomes. 05** **g) VIVA 05**					
				TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

9. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials (Any four)
01	Design & Impement a R-triggering circuit for a half wave-controlled rectifier circuit for 24V ac supply. The SCR to be used has the following data. Igmin = 0.1mA, Igmax =12 mA, Vgmin= 0.6V, Vgmax = 1.5 V
02	Study Data sheet of given IGBT (FGA15N120 IGBT) & present the following 1. Explain Pindescription of IGBT 2. ListExplainthe features of given IGBT 3. List alternatives for given IGBT 4. Listapplications of IGBT 5. Rig up a circuit to turn on/off IGBT 6. Explain switching characteristics of given IGBT
03	Study given Standard Rectifier Module (VUO190-16NO7) & & present the following 1. List the features of given Rectifier module 2. Listapplications of given Rectifier module 3. Explain the different ratings of the Rectifier module. 4. Check its suitability for a given specification Study Industrial rectifiers 1) List Types of Industrial rectifiers and their applications 2) Typical specifications of commercially available rectifiers 3) Cooling methods employed in rectifiers
04	Study the Intelligent Power Module(STK551U3A2A-E) & present the following 1. Explain features of Intelligent power module 2. List applications of Intelligent power module 3. Explain rating of given IPM
05	Study MOSFET driver IC(IR2110/IR2113) & present the following 1.Typical Connection 2.Absolute Maximum Ratings. 3.Recommended Operating Conditions. 4.Functional Block Diagram. 5.Pin configuration.

10. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

11. SEE–Model Theory Question Paper

Program	ELECTRICAL & ELECTRONICS ENGINEERING		IV
Course Name	POWER ELECTRONICS		50
Course Code	25EE42I	Duration	90 Min

Note: Answer any one full question from each section. Each full question carries equal marks.

Q No	Questions	Cognitive Levels	Course Outcomes	Marks
	Section -1			
1	a)Draw General Block Diagram of Power Electronic system b)Explain V –I characteristics of power diode.		1	5 5
2	a)Explain transfer characteristic curve of MOSFET and list Application of MOSFET. b) Explain Static V-I characteristic curve of SCR,		1	5 5
	Section -2			
3	a)Select a suitable MOSFET for 12V INVERTER for a residential application. b)Explain the Necessity of driver circuit for MOSFET	L3 L2	2	5 5
4	a)Select suitable IGBT for a given application. b)Explain the operation of MOSFET as switch.	L3 L2	2	5 5
	Section -3			
5	a)Explain the operation of the R-C firing circuit with waveforms. b) Explain How SCR can be protected against overvoltage and over current		2	10
6	a)Explain Digital firing scheme with waveforms b)Explain the necessity of heat sinks for MOSFET and IGBT,		2	5 5
	Section -4			
7	Explain the operation of a full-bridge inverter with neat waveforms.	L2	3	10
8	Explain the operation of a single-phase dual converter with neat waveforms .	L2	3	10
	Section -5			

9	a)ExplainWorking principle of Buck converter.. b)Draw the block diagram of SMPS. c) List any two sources of harmonics.	L2 L2 L3	3	5 3 2
10	a)Explain the operation of step down DC Chopper- b)Draw the block diagram of online UPS c)List the applications of SMPS	L2 L2 L3	3	5 3 2

11. **CIE- Model Practice Question Paper**

Program	**ELECTRICAL & ELECTRONICS ENGINEERING**		**Semester**	**IV**
Course Name	**POWER ELECTRONICS**	**Course Code :25EE42I**	**Duration**	**180 min**
Questions			**CO**	**Marks**
Rig up a suitable circuit to fire the given SCR above 90 degrees.			**2**	**50**
Scheme of assessment **h)** Portfolio evaluation of practice sessions from(Graded excercises) (Week-01 to Week-13) 5 **i) Demonstrate the working condition of power electronics devices (MOSFET). 5** **j) Drawing of the Circuit diagram using the right symbols 5** **k) Build the circuit as per the circuit diagram 10** **l) Demonstrate the working of the circuit 15** **m) Document the necessary readings of the expected circuit outcomes. 05** **n) VIVA 05**				
			Total Marks	**50**

1) Signature of the Examiner **2) Signature of the Examiner**

12. Equipment/softwarelistwithSpecificationforabatchof30students

Sl.No.	Particulars	Specification	Quantity
01	DCRegulatedpowersupply	(0-300V,2A)	6
02	DCRegulatedDualpowersupply	(0-30V,2A)	6
03	CathodeRayOscilloscope	Dualtrace,25 MHz.	6
04	DigitalMultimeter	31/2”1-ph	6
05	1-phInductionMotor	220v,1ph,	1
06	Battery	6V/12V60AH	3
07	SMPS		6
08	UPS2kVA	2kVA	2
Software			
01	GNU-Octave/MatLab/P-spice/Multisim		30 License

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	**ELECTRICAL & ELECTRONICS ENGG**	**Semester**	**IV**
Course Name	FUNDAMENTALS OF AUTOMATION TECHNOLOGY	**Type of Course**	Integrated
Course Code	**25EE43I**	**Contact Hours**	7Hrs/week 91Hrs/sem
Teaching Scheme	L: T:P 3:0:4	**Credits**	5
CIE Marks	**50**	**SEE Marks**	**50 (Practice)**

1. Rationale:

2. Course Outcomes : At the end of the Course, the student will be able to:

CO-01	Install configure Arduino IDE and write,debug and execute Arduino programs
CO-02	Design, build, and program a simple real life application using a Arduino Uno
CO-03	Select a suitable sensor and actuator for a given automation application and demonstrate its use in a specific application.
CO-04	Develop ladder diagrams for a given application and explain its implementation process using PLC

3. Course Content

WEEK	**CO**	**PO**	**Theory**	**Practice**
1	**1**		**Introduction to microcontroller**s, comparison between microprocessor and micro controller. Examples of microcontrollers in everyday life . what is arduino , Development board .	● **Tour of Arduino board** ● Install Arduino software ● **Arduino IDE: Installing the Arduino IDE. Overview of menus and tools in the IDE. Writing, compiling, and uploading code.** ● **uploading the first program ("Blink an LED").**
			● Tour of Arduino board. ● Arduino family : Uno and Leonardo,Mega and Due,The Micro and Small Arduino Boards,Yun & Lilypad	
			C Language Basics: ● What Is Programming Language**?**	

			• Arduino Program structure (https://docs.arduino.cc/learn/starting-guide/getting-started-arduino/) • Importance of setup() and loop(). • built-in functions: pinMode and digitalWrite • write a program to Blink an LED using above functions.	
2	1		• **Variables** • **Data types: Int , float , char, boolean** • Conditional operators. • Explain If statement • Explain for loop and While loop (https://docs.arduino.cc/language-reference/en/structure/control-structure/for/) **Function :** • What is a function • Function Parameters • return values • Global, Local, and Static Variables **Digital I/O Operations:** Digital Input and Output: Understanding digital signals (HIGH/LOW). Configuring pins as input or output (pinMode() function). Using digitalWrite() to control outputs and digitalRead() to read inputs.	1.write a program to illustrate use of If command 2. write a program to illustrate use of for loop and while loop. (https://docs.arduino.cc/built-in-examples/control-structures/ForLoopIteration/)
3	1		**Pull-Up and Pull-Down Resistors: What are pull-up and pull-down resistors? Preventing floating states in digital inputs** **Analog Signals and PWM:** Analog Input: Differences between digital and analog signals. Using analogRead() to read values from sensors. Voltage division and interpreting analog inputs (0–1023 scale). **Pulse Width Modulation (PWM): o Basics of PWM: Duty cycle and frequency. Applications: LED dimming and motor speed control. Using analogWrite() for PWM output.**	**1. Interfacing push buttons and LEDs to control ON/OFF states.** 2.Dimming an LED with a potentiometer using analogRead() (https://docs.arduino.cc/built-in-examples/analog/AnalogInput/) 3.Control the brightness of an LED using PWM. (https://docs.arduino.cc/built-in-examples/analog/Fading/)
4	2		**Introduction to Sensors: Types of sensors: Analog vs. digital. How sensors interact with Arduino Uno.** **Commonly Used Sensors: Temperature sensors (DHT11, LM35). Light sensors (LDR). Motion sensors (PIR).**	**Interfacing an LDR to measure light intensity.** (https://www.tinkercad.com/projects/Light-Sensor-Photoresistor-Arduino-Tinkercad) **2. Building a motion detection**

			Data Interpretation: Understanding sensor output. Calibrating sensors for accurate reading.	**system using a PIR sensor** (https://www.tinkercad.com/projects/PIR-Motion-Sensor-With-Arduino-in-Tinkercad)
5	**2**		Introduction to Actuators: What are actuators? Types: Servo motors, DC motors, and stepper motors.	**Controlling a servo motor to create basic robotic motion.**
			Servo Motors: How servo motors work (control signal and position). Applications of servo motors in robotics.	**2. Using a relay to switch a small DC motor ON/OFF.**
			Relay Modules: Basics of relays and their use in controlling high-power devices. Safety precautions when using relays	
6	**2**		**1.Serial Communication: o Introduction to serial communication. Using Serial.begin(), Serial.print(), and Serial.read()..** (https://docs.arduino.cc/language-reference/en/functions/communication/serial/begin/)	1. Sending and receiving data through the Serial Monitor. 2. Interfacing an I2C LCD module to display data.
	2		**2. I2C Protocol: o Basics of I2C communication (master-slave model). o Applications: Interfacing LCDs and EEPROMs.**	
	2		**3. SPI Protocol: Basics of SPI communication. applications: Interfacing SD cards and sensors. 4. Debugging with Serial Monitor: Using the Serial Monitor for debugging and data visualization**	
7	**3**		**1.Sensors: operation, characteristics and application: Inductive Proximity Sensors, Magnetic Proximity Sensor**	1a. Identify and test different sensors. **1b. Select a suitable proximity sensor for a given application and wire up the same. Ref 7(2)**
			2. Capacitive proximity sensors, Optical proximity sensors, Ultrasonic proximity sensors	**2.Develop a relay-based motor control automation such that the motor reverses its direction when the limit switches are activated.**
			3. Linear Position sensors, Photoelectric sensors. Infrared sensors Limit Switches.	
8	**3**		1. Inductive linear transducer, area sensors, flow sensors **Ref.7(1)**	1.Identify and test different sensors.

			2. Temperature sensors, colour sensors, Hall effect sensor 3a. Pressure sensors -Electronic pressure switches with binary output signal. 3b. Concept of Sensor latching. Latching Digital Hall effect sensor.	**2. Simple Hall effect sensor Latching ON/OFF Relay switch Circuit.**
9	**4**		1. **Fundamentals of control technology**: Meaning of control system, open loop and closed system with examples. Different types of controllers (PLC, CNC, Hard-wired programmed control systems, robot controllers)	**1.Demonstrate open loop and closed systems observed in everyday life**
			2. How programmable logic controllers (PLCs) work and their structure. Advantages and Disadvantages of PLC. Mathematical fundamentals – basic logic functions-Identity (YES function), Negation (NOT function), Conjunction (AND function), Disjunction (OR function), XNOR and XOR. 3. Examples of controller structure.	
10	**4**		1. **Programmable logic controllers-** Internal architecture and functional structure. Input/output modules.	1.a. Identify Components of PLC b. Identify different types of PLC c. Identify different input and output devices of PLC d. Identify the wiring mode of PLC-sourcing and sinking modes
			2a. List input / output devices of PLC. -List types of PLC. 2b. Functions of Programming equipment (Programmer/monitor) 3a. Explain PLC Programming Languages – Ladder diagram/ Functional Block Diagram /Instruction List/structured text. 3b. Explain scope of f IEC standard for PLC: IEC 61131	2.Identify and Install Programming Software and communication driver.
11	**4**		a. **Operation cycle of PLC**:1 Input scan, Program scan and Output scan. **1b. Operation modes of PLC**: program, run and test modes. Data files and program files.	. Develop and test the ladder programs for the following motor controls:) Starting from two different locations (OR Function)) Stopping from one position (NOT Function) c) Two hand operation (AND Function) d) Stopping from two different locations (NOT+OR or NOR Functions) e) Stopping if both signals are given (NOT+AND or NAND functions)) Memory function (Signal is maintained or holding) **Interlocking protection**

				(XNOR/XOR)
			2.Configuration of I/Os and Addressing I/Os, study of PLC symbols. 3. Procedure for drawing ladder diagram, connection of inputs and outputs to input and output module and entering ladder program into PLC (CPU). **Draw the ladder diagram for a simple example of one-contact, one-coil circuit and connection diagram showing how inputs are connected to the input and output module of PLC.**	2.Develop and test the ladder program for interlocking two motors, using **PLC simulation software**
12	**4**		1.Most commonly used PLC programming instructions and their applications: XIC, XIO, OTE. Latch, Unlatch	1a. Develop and test ladder program for switching ON motor 1, motor 2 and motor 3 in sequence with some time delay, using **PLC simulation software.** **1b. Develop and test the ladder program of the Alarm system for the following conditions: If one input is ON- nothing happens, if any two inputs are ON- a red light turns ON, If any three inputs are ON- a Hooter/Alarm turns ON, using PLC simulation software**
			2.Describe Timer instructions and their application: Describe Timer On Delay (TON), Timer Off Delay (TOF), Retentive Timer On (RTO) 3.Describe counter instructions and their application: COUNT UP, COUNTDOWN, UP/DOWN COUNTER Examples of use of counter and timer instructions.	2a. Develop and test ladder Program for fully Automatic Star-Delta starter, using **PLC simulation software.** **2b. Develop and test ladder Program to control automatic washing machine, using PLC simulation software.**
13	**4**		Wiring sensors to PLC **Wiring push button to PLC, and selector switch to PLC**	1. Develop and run simple Ladder programs to read sensor status and to control various output. **LED is turned ON when a (proximity sensor) sensor is activated.** i. Draw the ladder diagram ii. Draw PLC wiring diagram. iii. Wire push buttons to input module and LED to output module. iv. Enter the ladder program

				into the PLC simulator and execute. v. If the program is error free, Upload the program into PLC and execute. observe the output
			2.Wiring NPN sensor to PLC	
			3.Wiring PNP sensor to PLC	

4. References:

1.Massimo Banzi, "Getting Started with Arduino: The Open Source", Shroff Publishers & Distributors Pvt Ltd, 2014

2. Simon Monk, "Programming Arduino: Getting Started with Sketches", McGraw-Hill Education, Second Edition,2016

Margolis, "Arduino Cookbook", Shroff/O'Reilly Publication, 2nd edition 2012S

https://arduino-doc.readthedocs.io/en/latest/4.Questions%20and%20Answers/Questions%20and%20Answers/

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

6. SEE – Practice Assessment Methodologies

Sl. No	SEE – Practice Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Practice	180	50	20

7. CIE Theory Test model question paper

Program	**ELECTRICAL & ELECTRONICS ENGG**			**Semester - IV**	
Course Name	FUNDAMENTALS OF AUTOMATION TECHNOLOGY			**Test**	**I**
Course Code	**25EE43I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	a)What are the advantages in using Arduino over other microcontroller platforms? b)List the applications of microcontroller c)Which microcontroller is used in Arduino uno? What is the clock speed of this microcontroller? d) Explain the analogRead() function. e) Write a program to turn on built in LED of the arduino Uno for 1 sec and off for 1 sec		L2 L1 L2 L2 L3	5 5 5 5 5 5
2	a)Explain Digital write and Digital b)List features of arduino uno c)Explain the pinMode() function. d) What is an Arduino code called? What are the two major sections of an Arduino code? e) What does the syntax digitalWrite(13, HIGH) describe?		L2 L1 L2 L2 L3	5 5 5 5 5
	Section - 2			
3	a)**What is Pulse Width Modulation?** b)**What does the command analogWrite(127) mean?** c) Explain the main components of an Arduino microcontroller and their functions. d)**Explain syntax of Do while loop**		L2 L2	5 5 10

4	a)**Explain the delay() function** b). Describe the process of using Pulse Width Modulation (PWM) to control the speed of a motor or the brightness of an LED in an Arduino project. c)Describe the process of reading analog signals through an Arduino d) **Explain syntax of for loop**		L2 L2 L2 L2	5 5 10 5 5
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

8. CIE Practice Test model question paper

Program	**ELECTRICAL & ELECTRONICS ENGG**	**Semester**	**IV**
Course Name	FUNDAMENTALS OF AUTOMATION TECHNOLOGY	**Test**	**II**
Course Code	**25EE43I**	**Marks**	**50**
Name of the Course Coordinator:			
Questions		**CO**	**Marks**
write a program to blink a LED with delay of 5 millisecs			**50**
Scheme of assessment **a)** Portfolio evaluation of practice sessions (Week-01 to Week-13) 05 **b) Writing, compiling and debugging a program. 10** **c) uploading the and executing arduino program 10** **d) Demonstrate desired output from Arduino 20** **e) VIVA -VOCE 05**			
		TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

9. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials (Build any two one from Arduino and one from PLC)
01	Building Sun Tracking Solar Panel using an Arduino
02	How Does a Soil Moisture Sensor Work and How to use it with Arduino?
03	Measuring Turbidity of Water to Determine Water Quality using Arduino and Turbidity Sensor
04	Build Arduino DC Motor Speed Control using MOSFET Transistor.
05	Develop and implement PLC solution for automatic garage door opening.
06	Build PLC Based Automatic Alaram Systems in Plants(Indication, Enunciation,message)
07	Build PLC Based Automatic Counting System

10. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

11. SEE- Model Practice Question Paper

Program	**ELECTRICAL & ELECTRONICS ENGG**		**Semester**	
Course Name	FUNDAMENTALS OF AUTOMATION TECHNOLOGY	**Course Code 20EE42I**	**Duration**	**180 min**
Questions			**CO**	**Marks**
PART A **Building a motion detection system using a PIR sensor and Arduino uno** **PART B**				**50**

Develop and test ladder Program to control automatic washing machine, using PLC simulation software and upload ladder diagram and test the logic.		
Scheme of assessment **f)** Portfolio evaluation of practice sessions (Week-01 to Week-13) 5 **g) Writing, compiling and debugging a Arduino program. 5** **h) uploading the and executing arduino program and Demonstrate desired output from Arduino 5 +10 =15** **i) Draw the ladder diagram and PLC wiring diagram using right symbols . 5+5= 10 and simulate the ladder diagram usingPLC simulator** **j) Upload the Ladder program into PLC and execute and demonstrate 5+5=10 desired output from PLC.** **k) VIVA -VOCE 05**		
	Total Marks	**50**

1) Signature of the Examiner **2) Signature of the Examiner**

12. Equipment/software list with Specification for a batch of 30 students

Sl.No.	Particulars	Specification	Quantity
01	Arduino UNO or Compatible Board (1 no.)		30
02	Breadboard (1 no.)		30
03	USB Power Cable		30
04	Linear Actuator Solenoid 12V		30each
05	Inductive. Proximity Sensors. Magnetic Proximity Sensor, Limit Switches		30each
06	Capacitive proximity sensors, Optical proximity sensors Ultrasonic proximity sensors		30each
07	Infrared sensors, Pressure Sensor and Switch		30each
08	Inductive linear transducer , Area sensors, Flow sensors, Temperature sensors, colour sensors		30each
09	Inductive linear transducer , Area sensors, Flow sensors, Temperature sensors, colour sensors		30each
10	PLC Systems with digital I/P, O/P modules and software	12/24v Dc/relay, 8 Digital Inputs, 4 Digital Outputs, ethernet card standard micro SD card, integrated webserver	15
11	1 HP induction motor with DOL starter	1 HP	15

Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	ELECTRICAL AND ELECTRICAL ENGINEERING	**Semester**	**IV**
Course Name	ELECTRICAL POWER SYSTEM DESIGN	**Type of Course**	Integrated
Course Code	**25EE44I**	**Contact Hours**	7 hours/week 91 hours/semester
Teaching Scheme	L: T:P::3:0:4	**Credits**	5
CIE Marks	**50**	**SEE Marks**	**50 (Practice)**

1. Rationale:

The course prepares students to meet the demands of the electrical engineering industry by providing them with the skills to create, analyze, and optimize electrical designs using CAD tools. This practical, hands-on experience ensures that students are well-equipped to handle real-world electrical engineering challenges and contribute effectively to the industry.

2. Course Outcomes:At the end of the Course, the student will be able to:

CO-01	Students will learn to use CAD software for creating simple electrical diagrams and layouts, such as residential wiring and LT distribution panels.
CO-02	Students will be able to design and draft substation layouts, motor control panels, and AMF panels using CAD tools.
CO-03	Students will gain the ability to convert 2D electrical drawings into 3D models, specifically for components like the Squirrel Cage Induction Motor.
CO-04	Students will understand the basics of integrating PLC modules into electrical designs using CAD software.

3. Course Content

WEEK	CO	PO	Theory	Practice
1	1	1,4,7	1.introduction to CAD ● Explain the significance of CAD in electrical engineering. ● What are the benefits of using CAD for electrical design compared to manual drafting? ● List and explain different CAD file formats used in electrical drafting.	● Practice the basic CAD commands. ● commands. Practice Essential commands like–line types, line weight, scale, unit, ● Layer,block,insert,explode, purge, table, attribute, quick select ● view,multi-view,break,join, filter, find, pan, list match properties and related commands **Ref.4(1,2,3,4,5)**
			● What is the function of Common CAD commands ? ● Describe the function of the following commands in CAD: Line, Polyline, Trim, and Mirror.	
			● What is the use of 'Layer' in CAD, and why is it important to manage layers effectively in electrical drawings? ● Describe the procedure for inserting	

			electrical symbols from a CAD library into a drawing	
2	1,2	1,4,7	Substation ● Explain the components of a **110 KV/11KV** substation. Why is it important to create a single-line diagram for substations? ● Describe the difference between a **110KV/11KV** and **11KV/440V** substation	1. Single line diagram of 110 KV/11KV MUSS.Ref.4**(6,7)** **2.** Single line diagram of 11 KV/440V substation-**Ref.** 4**(8)**
3	1,2	1,4,7	Electrical Wiring of a residential The figure gives the plan of residence. It is to be wired up with AEH installation. draw the wiring plan and indicate all the fixtures. **Ref.**4**(2) (Example 1 page no 253)** ● find the number of outlets and number of circuits ● Create BOM (Bill of material) 2.Draw and Create BOM(Bill Of Material): Electrical wiring of a small workshop **Ref.**4**(2) (Example 8 page no 263)**	1.Draw the wiring layout of residential building -**Ref.**4**(2)** 2) Draw the layout diagram for distribution of power supply to a polytechnic having the following loads at 400/230V, 50Hz with 4 wire, A,C system **Ref.**4**(2) (Example 13 page no 268)**
4	1,2	1,4,7	Design a main LT distribution panel for the following load requirements in a polytechnic lecture Hall - 2KW Machine Shop-10 HP load, Computers - 20 nos Office lighting – 1 kw, MCB, MCCB, Busbar, cables (solve minimum 2 problems)	Draw the SLD for LT and GA Diagram distribution panel and mention ratings of busbar MCCB,MCB and cables
5	1,2	1,4,7	Motor control panel wiring. Detailed study of motor control circuits: DOL and star-delta starters, reversing circuits Design an MCC panel for the following load ● Machine Shop-7.5 HP load, ● Lift -5HP Find Busbar size, MCCB, MCB, and starter (Contacors)	1 To design a power circuit and control circuit 5 HP Induction motor **Ref.**4**(9)** 2.To design a power circuit and control circuit Induction motorin CAD **-Ref.**4**(10)** (Select suitable starters for given above loads)
6	1,2	1,4,7	Two-Speed, One-Direction, 3-Phase Motor Power & Control Diagrams **Ref.**4**(2,11)** ● Components Needed ● Power Diagram ● Schematic Wiring Diagram ● Control Wiring Diagram ● Control Circuit Diagram	Two-Speed, One-Direction, 3-Phase Motor Power & Control Diagrams - **Ref.**4**(2,11)**
7	1,2	1,4,7	Design a simple AMF panel- select the switchgear and its ratings as per load requirement. Design practice using CAD. a simple AMF panelGenerate the BOM	Design a simple AMF panel- Prepare the GA diagram, SLD, and control wiring drawings as per standard practice using CAD.
8			Electrical DC Lap winding DC Lap winding	Designing a single-layer lap winding diagram for a DC Machine using

	1,2	1,4,7	Introduction	CAD Ref.4(12) Designing a duplex lap winding diagram for a DC Machine using CADRef.4(13)
			Classification of windings in DC Machine ● Single Layer Winding.. ● Two Layer Winding.	
			● Lap Winding design procedure ● Solved problems	
9	1,2	1,4,7	Developed Winding Diagram Of 1-ph, AC Induction Motor	Develop a winding diagram for a 1-ph, Induction Motor, make terminal connections for Running & Starting Winding.
10	3	1,3,4,7	Developed Winding Diagrams of 3-ph A.C.Machines: Single Layer Double Layer	1. Draw a developed winding diagram for an AC machine having single layer wave wing **-Ref.4(2) (Example 13 page no 170)** 2. Draw a developed winding diagram for an AC machine having double layer wave wing **-Ref.4(2) (Example 13 page no 170)**
11	3	1,3,4,7	Developed Winding Diagrams of 3-ph A.C.Machines: Single Layer Double Layer	Draw a developed winding diagram for an AC machine having single layer lap wing **-Ref.4(2) (Example 13 page no 162)** Draw a developed winding diagram for an AC machine having single layer Lap wing **-Ref.4(2) (Example 13 page no 174)**
12	4	1,3,4,7	2D to 3D Drawing-Squirrel Cage Induction motor.	View Showing Different Parts. **Ref.4(14)**
13	4	1,3,4,7	PLCModule	Design a Simple PLC Module showingI/Points.Ref.4**(15)**

4. References:

1. Computer Aided Electrical Drawing - YOGESH, NAGARAJA,NANDAN PHI Publication
2. ElectricalDrafting-S.F.DEVALAPUR
3. https://www.youtube.com/watch?v=pvKVy-eMDYc
4. https://www.youtube.com/watch?v=2ni0AWbloQA
5. https://www.youtube.com/watch?v=wIN61lmZByw
6. https://www.youtube.com/watch?v=YXLhvA7dMb4
7. https://youtu.be/ZRXlWoT-FRU?si=jrWoG60jZ4VSWKni
8. https://youtu.be/WkrJXXex1AE?si=gBxrXWEJPBcXLRru
9. https://youtu.be/z91gWVhLJeQ?si=PpY--kTVRoup8DZH
10. https://youtu.be/bg-UEF-N5ss?si=1so3jcgcmfpX1Q2n
11. https://www.electricaltechnology.org/2014/06/2-speeds-1-direction-3-phase-motor.html
12. **https://youtu.be/ZxACANKAMEc?si=pZNqHwr1BYl6Sdf1**
13. **https://youtu.be/NUvFeEKVO8g?si=HU38OxRRM0fKuwcm**
14. https://youtu.be/fAN9jxydoMA?si=cehyUwO_DwNR_8eh
15. https://youtu.be/9l1_4WD2R-Y?si=7uBCOhaPHZVgLNNi

5. CIE Assessment Methodologies

Sl. No	CIE Assessment	Test Week	Duration (minutes)	Max marks	
1.	CIE-1TheoryTest	4	90	50	Average of all CIE=50 Marks
2.	CIE-2Practice Test	7	180	50	
3	CIE-3TheoryTest	10	90	50	
4.	CIE-4Practice Test	13	180	50	
5	CIE-5 Portfolio evaluation of all the activities through Rubrics	1-13		50	
				Total	**50 Marks**

6. SEE – Practice Assessment Methodologies

Sl. No	SEE – Practice Assessment	Duration (minutes)	Max marks	Min marks to pass
1.	Semester End Examination-Practice	180	50	20

7. CIE Theory Test model question paper

Program	ELECTRICAL AND ELECTRICAL ENGINEERING			**Semester - IV**	
Course Name	ELECTRICAL POWER SYSTEM DESIGN			**Test**	**I/III**
Course Code	**25EE44I**	**Duration**	**90 min**	**Marks**	**50**
Name of the Course Coordinator:					
Note: Answer any one full question from each section. Each full question carries equal marks.					

Q.No	Questions	Cognitive Level	Course Outcome	Marks
	Section - 1			
1	a) Explain the significance of CAD in electrical engineering b) Describe the function of the following commands in CAD: Line, Polyline, Trim, and Mirror c) Explain the components of a **110 KV/11KV** substation. Why is it important to create a single-line diagram for substations?	L2/L3	1,2	5+10+10 = 25
2	a) List and explain different CAD file formats used in electrical drafting	L2/L3	1,2	5+10+10 = 25

	b) Describe the procedure for inserting electrical symbols from a CAD library into a drawing c) Describe the difference between a **110KV/11KV** and **11KV/440V** substation			
Section - 2				
3	The figure gives the plan of residence. It is to be wired up with AEH installation. draw the wiring plan and indicate all the fixtures. ● find the number of outlets and number of circuits ● Create BOM (Bill of material	L3/L4	1,2	15+10=25
4	Draw GA diagram & SLD diagram of an LT panel for a factory with following loads: ● Machine shop - 3 ph 415 V 10 Hp load ● Borewell pump-3 ph 415 V 5 Hp x 2 Nos. ● Office with 20 computers ● Office lighting-1 KW Assume cos 0.8 select suitable cables, Wires. Switch gears MCB. Class. Size Breaking Capacity,. Draw SLD and Mark the dimensions of the panel.	L3/L4	1,2	15+10=25
Note for the Course coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.				

Signature of the Course Coordinator **Signature of the HOD** **Signature of the IQAC Chairman**

8. CIE Practice Test model question paper

Program	ELECTRICAL AND ELECTRICAL ENGINEERING			**Semester**	**IV**
Course Name	ELECTRICAL POWER SYSTEM DESIGN			**Test**	**II/IV**
Course Code	**25EE44I**	**Duration**	**180 min**	**Marks**	**50**
Name of the Course Coordinator:					
Questions				**CO**	**Marks**
1. Single line diagram of 110 KV/11KV MUSS.Ref.4**(6,7)** 2. Single line diagram of 11 KV/440V substation-**Ref.** 4**(8)** **3.** Draw the wiring layout of residential building -**Ref.**4**(2)** **4.** Draw a wiring layout of a small workshop with lathes,1 drilling machine,1 welding machine,1 grinding machine, and generate BOM (Bill of Material). 5. Draw the layout diagram for distribution of power supply to a polytechnic having the following loads at 400/230V, 50Hz with 4 wire, A,C system **Ref.**4**(2) (Example 13 page no 268)** 6. To design a power circuit and control circuit 5 HP Induction motor using DOL **Ref.**4**(9)** **7.** To design a power circuit and control circuit Induction motor using a star-delta starter circuit in CAD **-Ref.**4**(10)** **8.** Two-Speed, One-Direction, 3-Phase Motor Power & Control Diagrams -				**1,2**	**50**

Ref.4(2,11) **9.** Design a simple AMF panel- Prepare the GA diagram, SLD, and control wiring drawings as per standard practice using CAD		
Scheme of assessment **a) Use of suitable commands** **b) Labeling** **c) Indexing**		 **25** **15** **10**
	TotalMarks	**50**

Signature of the Course Coordinator **Signature of the HOD**

9. Suggestive Activities for Tutorials:

The List is an Example and not inclusive of all possible activities of the course. Student and Faculty are encouraged to choose activities that are relevant to the topic

Sl.No.	Suggestive Activities for Tutorials
01	Report and function of components Draw the SLD for a **110 KV/11KV MUSS (Main Unit Substation)**. ● Include components like transformers, circuit breakers, and isolators.
02	Report and function of components Create the SLD for an **11 KV/440V substation**. ● Add distribution transformers, feeders, and load representations.
03	**Objective:** Design wiring layouts and power distribution diagrams. **Activities:** 1. Draw the wiring layout for a **residential building**. ○ Include power points, switches, and lighting circuits. 2. Design the power supply layout for a **polytechnic institute**. ○ Loads include equipment, labs, classrooms, and admin blocks. ○ Apply proper load distribution techniques.
04	**Objective:** Create LT and GA distribution panel diagrams. **Activities:** 1. Draw the **SLD for the LT Distribution Panel**. 2. Create the **GA Diagram for the Distribution Panel**, including ratings of: ○ Busbars, MCCBs, MCBs, and cables. 3. Annotate all components with proper symbols and specifications.
05	Motor Power and Control Circuits **Objective:** Design motor control circuits for various applications. **Activities:** 1. Design a **power and control circuit for a 7.5 HP Induction Motor**. ○ Choose appropriate starters and overload protection.
06	PLC Module Design **Objective:** Create and simulate basic PLC modules. **Activities:** 1. Design a **simple PLC module** showing input/output points. ○ Include ladder logic diagrams for automated systems.

10. Rubrics for Assessment of Activity(Qualitative Assessment)

Sl. No.	Dimension	Beginner	Intermediate	Good	Advanced	Expert	StudentsScore
		2	4	6	8	10	
1		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	8
2		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	6
3		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
4		Descriptor	Descriptor	Descriptor	Descriptor	Descriptor	2
	Average Marks=(8+6+2+2)/4=4.5						**5**

Note: Dimension and Descriptor shall be defined by the respective course coordinator as per the activities

11. **SEE- Model Practice Question Paper**

Program	ELECTRICAL AND ELECTRICAL ENGINEERING		**Semester**	**IV**
Course Name	ELECTRICAL POWER SYSTEM DESIGN	**Course Code** **25EE44I**	**Duration**	**180 min**
Questions			**CO**	**Marks**
1. Single line diagram of 110 KV/11KV MUSS.Ref.4**(6,7)** 2. Single line diagram of 11 KV/440V substation-**Ref.** 4**(8)** 3. Draw the wiring layout of residential building -**Ref.**4**(2)** 4. Draw a wiring layout of a small workshop with lathes,1 drilling machine,1 welding machine,1 grinding machine, and generate BOM (Bill of Material). 5. Draw the layout diagram for distribution of power supply to a polytechnic having the following loads at 400/230V, 50Hz with 4 wire, A,C system **Ref.**4**(2) (Example 13 page no 268)** 6. To design a power circuit and control circuit 5 HP Induction motor using DOL **Ref.**4**(9)** 7. To design a power circuit and control circuit Induction motor using a star-delta starter circuit in CAD **-Ref.**4**(10)** **8.** Two-Speed, One-Direction, 3-Phase Motor Power & Control Diagrams - **Ref.**4**(2,11)** **9.** Design a simple AMF panel- Prepare the GA diagram, SLD, and control wiring drawings as per standard practice using CAD 10. Designing a single-layer lap winding diagram for a DC Machine using CAD Ref.4(12 11. Designing a duplex lap winding diagram for a DC Machine using CAD Ref.4(13) 12. Develop a winding diagram for a 1-ph, Induction Motor, make terminal connections for Running & Starting Winding 13. Draw a developed winding diagram for an AC machine having single layer wave wing **-Ref.**4**(2) (Example 13 page no 170)** 14. Draw a developed winding diagram for an AC machine having double layer wave wing **-Ref.**4**(2) (Example 13 page no 170)** **15.** Draw a developed winding diagram for an AC machine having single layer lap wing **-Ref.**4**(2) (Example 13 page no 162)** **16.** Draw a developed winding diagram for an AC machine having single layer			**50**

Lap wing -**Ref.**4**(2) (Example 13 page no 174)** **17. 3D view showing different parts** **18. Design a Simple PLC Module showing I/O points**		
Scheme of assessment **a) Use of suitable commands** **b) Labeling** **c) Indexing**		 **25** **15** **10**
	Total Marks	**50**

1) Signature of the Examiner **2) Signature of the Examiner**

12. Equipment/software list with Specification For Batch Of 30 Students

Sl.No.	Particulars	Specification	Quantity
01	Personal Computer	● Operating System: 64-bit Microsoft Windows 10 ● Processor: 2.5 GHz (3+ GHz recommended) ● Memory: 8 GB (16GB recommended) ● Disk space: 1TB. ● Display: 1920 x 1080 resolution with True Color.	20
	Electrical Computer Aided Drafting Software/ AutoCAD Electrical 2021/Proficad	Student edition	20